
Predicting Future CO2 Emission Trends

**ISM 645 / Predictive
Analytics**

**By:
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Context & Objective

- The annual rate of increase in atmospheric carbon dioxide over the past 60 years is about 100 times faster than previous natural increases, such as those that occurred at the end of the last ice age 11,000-17,000 years ago.. - [Climate.gov](https://climate.gov)
- Using historical data from 1995 to 2018, we aim to predict future CO2 emissions for individual countries and at the global level



Data Set Overview

- We used a data set from Kaggle that shows a numerical value of the millions of metric tons of CO2 per industry, per country, from the years 1995 - 2018.

<https://www.kaggle.com/dysfunctionalhuman/global-co2-emissions>

- 66 total countries
- 135 industries per country
- 1995 to 2018

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
	ObjectID	Country	ISO2	ISO3	Indicator	Unit	Source	CTS_Code	CTS_Name	CTS_Full_De	Industry	Scale	F1995	F1996	F1997	F1998	F1999	F2000
1	1	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Accommodation and food services	Units	0.159	0.158	0.163	0.177	0.175	0.183
2	2	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Activities of households as employers; un	Units	NaN	NaN	NaN	NaN	NaN	NaN
3	3	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Administrative and support services	Units	0.326	0.351	0.363	0.36	0.336	0.338
4	4	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Agriculture, hunting, forestry	Units	8.914	9.58	9.625	9.18	9.124	9.02
5	5	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Air transport	Units	1.957	2.518	2.523	2.664	2.602	2.499
6	6	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Arts, entertainment and recreation	Units	0.073	0.077	0.081	0.074	0.07	0.069
7	7	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Basic metals	Units	4.043	4.908	5.167	5.697	5.573	5.906
8	8	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Chemicals and pharmaceutical products	Units	1.161	1.192	1.134	1.304	1.299	1.315
9	9	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Coke and refined petroleum products	Units	4.338	3.509	4.642	4.505	4.146	4.144
10	10	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Computer, electronic and optical produc	Units	0.01	0.011	0.011	0.011	0.009	0.008
11	11	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Construction	Units	0.268	0.306	0.306	0.445	0.35	0.313
12	12	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Education	Units	0.205	0.203	0.209	0.222	0.233	0.243
13	13	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Electrical equipment	Units	0.006	0.007	0.009	0.008	0.007	0.009
14	14	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Electricity, gas, steam and air conditionir	Units	18.652	21.82	20.189	22.337	27.149	25.805
15	15	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Fabricated metal products	Units	0.013	0.015	0.027	0.019	0.02	0.027
16	16	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Financial and insurance activities	Units	0.234	0.234	0.242	0.258	0.264	0.275
17	17	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Fishing and aquaculture	Units	0.103	0.122	0.132	0.129	0.111	0.108
18	18	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Food products, beverages and tobacco	Units	2.123	2.251	2.285	2.607	2.732	2.819
19	19	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Human health and social work activities	Units	0.436	0.462	0.467	0.466	0.412	0.385
20	20	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	IT and other information services	Units	0.016	0.016	0.016	0.026	0.026	0.027
21	21	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Land transport and transport via pipeline	Units	8.988	10.31	10.544	11.036	10.263	9.86
22	22	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Machinery and equipment n.e.c.	Units	0.028	0.032	0.046	0.036	0.033	0.036
23	23	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Manufacturing nec; repair and installatio	Units	2.157	1.972	2.601	2.08	1.592	1.379
24	24	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Mining and quarrying, energy producing p	Units	3.702	5.358	5.842	8.324	8.812	9.406
25	25	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Mining and quarrying, non-energy produc	Units	0.185	0.22	0.26	0.189	0.131	0.164
26	26	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Mining support service activities	Units	0.042	0.048	0.057	0.045	0.041	0.043
27	27	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Motor vehicles, trailers and semi-trailers	Units	0.198	0.201	0.244	0.242	0.191	0.187
28	28	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Other non-metallic mineral products	Units	2.367	2.287	2.62	2.763	2.419	2.123
29	29	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Other service activities	Units	0.182	0.188	0.196	0.193	0.183	0.181
30	30	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Other transport equipment	Units	0.01	0.01	0.012	0.013	0.01	0.011
31	31	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Paper products and printing	Units	0.585	0.624	0.625	0.615	0.521	0.576
32	32	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Pharmaceuticals, medicinal chemical an	Units	0.428	0.437	0.412	0.442	0.432	0.431
33	33	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Postal and courier activities	Units	0.042	0.046	0.049	0.05	0.046	0.043
34	34	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Professional, scientific and technical act	Units	0.396	0.43	0.465	0.429	0.362	0.336
35	35	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Public administration and defence; comp	Units	0.392	0.406	0.426	0.51	0.537	0.596
36	36	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Publishing, audiovisual and broadcasting	Units	0.128	0.136	0.141	0.155	0.137	0.128
37	37	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Real estate activities	Units	0.295	0.291	0.299	0.309	0.324	0.335
38	38	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Rubber and plastics products	Units	4.633	4.225	5.576	4.455	3.399	2.823
39	39	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Telecommunications	Units	0.102	0.102	0.105	0.108	0.109	0.112
40	40	Argentina	AR	ARG	CO2 emissio	Millions of Metric tons of CO2	OECD (2021)	ECNC	CO2 Emissio	Environment	Textiles, textile products, leather and foo	Units	1.845	1.989	2.574	2.294	1.972	2.32

Data Cleaning & Preparation

Datasets Used was divided into parts

CO2 Emissions, CO2 Intensities, CO2 Multipliers, and Clustering Data sheets

Covered **1995 to 2018** with country-level and industry-level data

Features Used: Emissions data, variance, intensities, and industry metrics

Challenges Encountered

Missing values and inconsistent data across years and countries

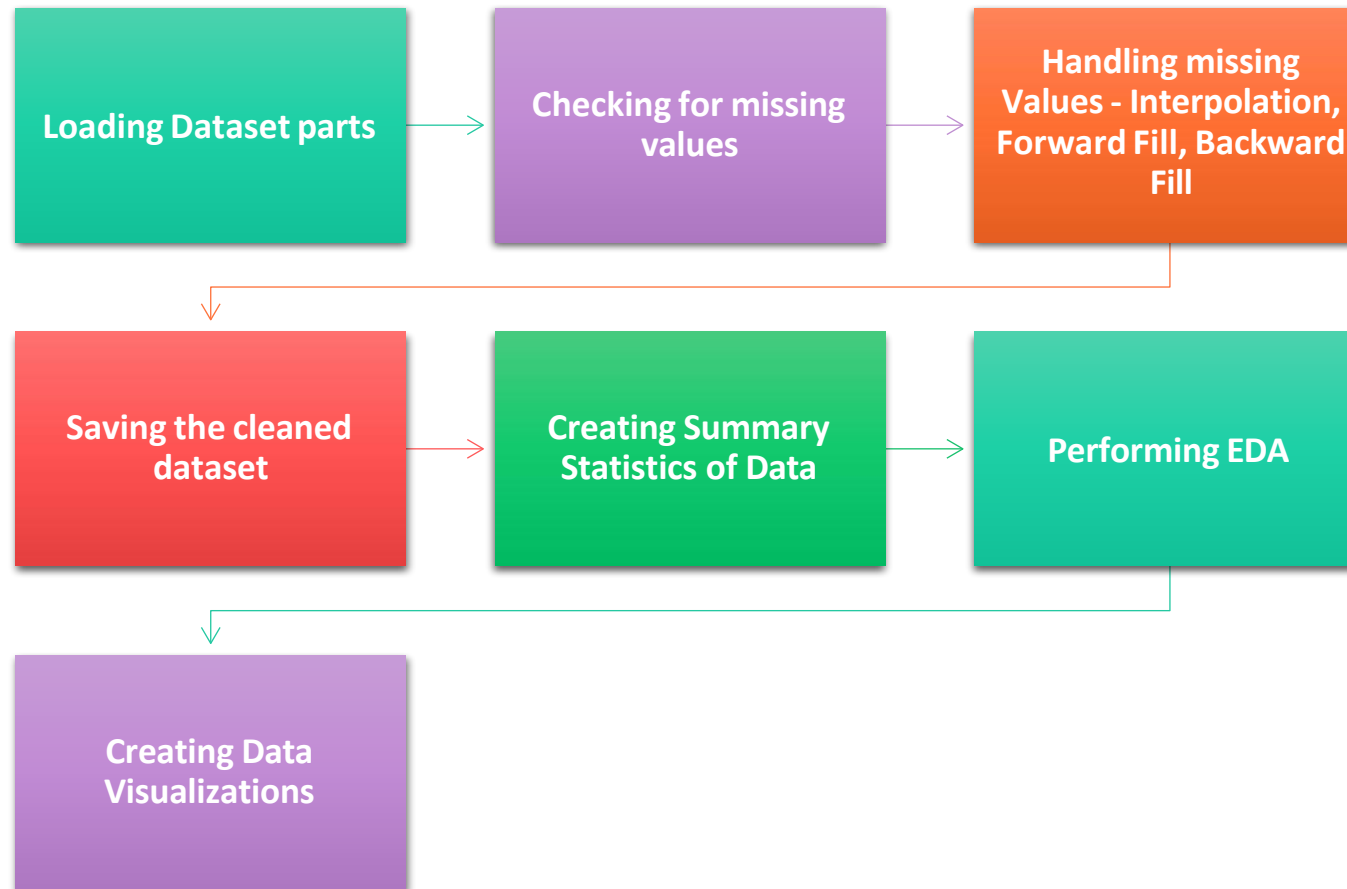
Outliers affecting statistical analyses

Datasets with varying formats and column structures

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Country	ISO2	ISO3	Indicator	Unit	Source	CTS Code	CTS Name	CTS Full Desc	Industry	Scale	F1966	F9999	F1967	F9999	F1968	F9999	F2000	F2001	F2002	F2003	F2004	F2005	F2006	F2007	F2008	
1 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Accommodation Units				0.159	0.158	0.168	0.17	0.175	0.183	0.175	0.195	0.208	0.21	0.21	0.21	0.21	0.21	0.21	
2 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Activities of Units				N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
3 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Administrative Units				0.326	0.351	0.363	0.36	0.336	0.339	0.366	0.365	0.412	0.416	0.41	0.41	0.41	0.41	0.41	0.41
4 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Agriculture Units				8.814	5.958	9.625	8.18	9.124	8.02	8.46	9.348	9.348	10.46	10.46	12.24	12.24	12.24	12.24	12.24
5 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Air transport Units				1.857	2.518	2.529	2.664	2.662	2.699	2.984	2.949	2.981	3.261	3.261	3.261	3.261	3.261	3.261	3.261
6 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Arts, entertainment Units				0.073	0.077	0.081	0.074	0.07	0.069	0.063	0.067	0.07	0.08	0.08	0.08	0.08	0.08	0.08	0.08
7 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Basic metals Units				0.043	4.908	5.167	6.697	5.575	5.906	5.064	5.302	5.794	6.4	6.4	6.4	6.4	6.4	6.4	6.4
8 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Chemicals Units				1.61	1.192	1.134	1.394	1.299	1.315	1.794	1.325	1.292	1.91	1.91	1.91	1.91	1.91	1.91	1.91
9 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Chemicals Units				4.358	3.049	4.642	4.935	4.146	4.144	4.641	4.603	4.957	5.07	5.07	5.07	5.07	5.07	5.07	5.07
10 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Chemicals Units				0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
11 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Construction Units				0.266	0.306	0.445	0.35	0.313	0.335	0.281	0.347	0.407	0.07	0.07	0.07	0.07	0.07	0.07	0.07
12 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Education Units				0.205	0.203	0.209	0.222	0.238	0.243	0.238	0.227	0.247	0.07	0.07	0.07	0.07	0.07	0.07	0.07
13 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Electronic Units				0.008	0.007	0.009	0.009	0.007	0.009	0.006	0.011	0.012	0.01	0.01	0.01	0.01	0.01	0.01	0.01
14 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Electronic Units				38.652	21.82	21.98	22.337	23.148	26.305	31.480	20.311	24.016	24.016	24.016	24.016	24.016	24.016	24.016	24.016
15 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Electronic Units				0.015	0.015	0.017	0.019	0.02	0.027	0.026	0.045	0.052	0.052	0.052	0.052	0.052	0.052	0.052	0.052
16 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Financial services Units				0.234	0.234	0.242	0.258	0.264	0.275	0.264	0.307	0.334	0.334	0.334	0.334	0.334	0.334	0.334	0.334
17 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Fishing and aquaculture Units				0.038	0.122	0.132	0.129	0.111	0.108	0.1	0.158	0.168	0.168	0.168	0.168	0.168	0.168	0.168	0.168
18 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Food products Units				2.123	2.251	2.285	2.607	2.732	2.819	2.790	2.872	2.609	2.86	2.86	2.86	2.86	2.86	2.86	2.86
19 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Food products Units				0.456	0.456	0.456	0.456	0.456	0.456	0.456	0.456	0.456	0.456	0.456	0.456	0.456	0.456	0.456	0.456
20 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Food and other products Units				0.016	0.016	0.016	0.026	0.026	0.027	0.025	0.028	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03
21 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Food and other products Units				0.068	10.31	10.544	11.036	10.263	9.86	9.124	8.827	10.282	11.036	11.036	11.036	11.036	11.036	11.036	11.036
22 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Machinery and equipment Units				0.028	0.032	0.046	0.036	0.033	0.036	0.029	0.036	0.036	0.036	0.036	0.036	0.036	0.036	0.036	0.036
23 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Manufacturing Units				1.167	1.872	2.401	2.38	1.842	1.376	1.446	1.666	1.89	1.89	1.89	1.89	1.89	1.89	1.89	1.89
24 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Manufacturing Units				0.319	0.642	0.319	0.642	0.319	0.642	0.319	0.642	0.319	0.642	0.319	0.642	0.319	0.642	0.319	0.642
25 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Mining and quarrying Units				0.165	0.22	0.26	0.169	0.131	0.164	0.11	0.203	0.207	0.207	0.207	0.207	0.207	0.207	0.207	0.207
26 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Mining and quarrying Units				0.042	0.048	0.067	0.045	0.041	0.043	0.038	0.056	0.059	0.059	0.059	0.059	0.059	0.059	0.059	0.059
27 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Motor vehicles Units				0.198	0.201	0.244	0.342	0.191	0.197	0.14	0.116	0.117	0.117	0.117	0.117	0.117	0.117	0.117	0.117
28 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Other non-ferrous metals Units				3.287	2.287	2.42	2.783	2.118	2.123	1.896	1.58	1.332	2.18	2.18	2.18	2.18	2.18	2.18	2.18
29 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Other non-ferrous metals Units				0.181	0.181	0.181	0.181	0.181	0.181	0.181	0.181	0.181	0.181	0.181	0.181	0.181	0.181	0.181	0.181
30 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Other transport Units				0.01	0.01	0.012	0.013	0.01	0.011	0.008	0.009	0.006	0.006	0.006	0.006	0.006	0.006	0.006	0.006
31 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Paper products Units				0.585	0.624	0.625	0.645	0.621	0.621	0.576	0.506	0.519	0.571	0.571	0.571	0.571	0.571	0.571	0.571
32 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Pharmaceuticals Units				0.428	0.437	0.412	0.442	0.432	0.431	0.623	0.469	0.460	0.460	0.460	0.460	0.460	0.460	0.460	0.460
33 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Pharmaceuticals Units				0.042	0.042	0.042	0.042	0.042	0.042	0.042	0.042	0.042	0.042	0.042	0.042	0.042	0.042	0.042	0.042
34 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Professional services Units				0.264	0.43	0.465	0.429	0.362	0.336	0.287	0.315	0.216	0.216	0.216	0.216	0.216	0.216	0.216	0.216
35 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Public administration Units				0.260	0.406	0.426	0.51	0.337	0.396	0.585	0.788	0.919	0.919	0.919	0.919	0.919	0.919	0.919	0.919
36 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Publishing and printing Units				0.128	0.136	0.141	0.155	0.137	0.128	0.111	0.113	0.111	0.111	0.111	0.111	0.111	0.111	0.111	0.111
37 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Real estate and construction Units				0.256	0.291	0.299	0.309	0.324	0.335	0.323	0.316	0.326	0.326	0.326	0.326	0.326	0.326	0.326	0.326
38 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Real estate and construction Units				0.425	0.425	0.425	0.425	0.425	0.425	0.425	0.425	0.425	0.425	0.425	0.425	0.425	0.425	0.425	0.425
39 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Telecommunications Units				0.102	0.102	0.102	0.108	0.108	0.112	0.116	0.116	0.116	0.116	0.116	0.116	0.116	0.116	0.116	0.116
40 Argentina	AR	ARG	CO2 emissions	Millions of Mt.OECD (2021)	EMC6	EMC6	CO2 Emissions Environment: Textiles, textile units Units				1.845	1.989	2.574	2.294	1.872	2.32	2.381	2.040	2.341	2.341	2.341	2.341	2.341	2.341	2.341	2.341

[illegible]

Data Cleaning...



Missing Values After Cleaning:

```
ObjectId      0
Year          0
Industry      0
Unit          0
Indicator     0
--
Tunisia       0
Turkey        0
United Kingdom 0
United States 0
Vietnam       0
Length: 71, dtype: int64
```

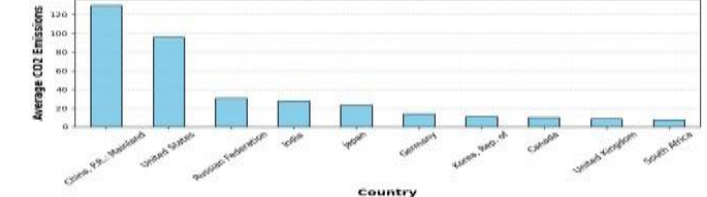
--- Missing Values in CO2 Emissions ---

```
ObjectId      0
Year          0
Industry      0
Unit          0
Indicator     0
--
Tunisia       24
Turkey        24
United Kingdom 24
United States 24
Vietnam       33
Length: 71, dtype: int64
Total missing values in CO2 Emissions: 4216
```

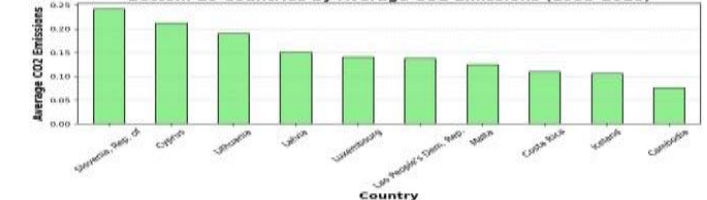
Summary Statistics:

	ObjectId	Year	Argentina	Australia	Austria
count	1000.00000	1000.00000	1000.00000	1000.00000	1000.00000
mean	540.50000	2006.50000	480.660208	641.656296	229.320557
std	311.91345	6.925393	729.319785	1589.353064	362.637795
min	1.00000	1995.00000	28.370798	44.151306	18.756068
25%	270.75000	2000.75000	136.375817	175.433333	52.710277
50%	540.50000	2006.50000	235.697142	314.352742	94.563840
75%	810.25000	2012.25000	480.721588	544.350116	226.954587
max	1000.00000	2018.00000	6429.313845	17690.233100	2993.007210

Top 10 Countries by Average CO2 Emissions (1995-2018)



Bottom 10 Countries by Average CO2 Emissions (1995-2018)



EDA (Exploratory Data Analysis)

```
for sheet_name, df in cleaned_data.items():
    print(f"\n--- EDA for {sheet_name} ---")

    # 1. Summary Statistics
    print("\nSummary Statistics:")
    print(df.describe()) # Summary for numeric columns

    # 2. Missing Values Verification
    print("\nMissing Values After Cleaning:")
    print(df.isnull().sum()) # Ensure no missing values remain
```

--- EDA for CO2 Emissions ---

Summary Statistics:

	ObjectId	Year	Argentina	Australia	Austria
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
mean	540.500000	2006.500000	2.419669	7.247410	1.079161
std	311.913145	6.925393	5.831435	27.523043	2.146356
min	1.000000	1995.000000	0.004000	0.024000	0.001000
25%	270.750000	2000.750000	0.146500	0.310250	0.154000
50%	540.500000	2006.500000	0.408000	0.753000	0.356500
75%	810.250000	2012.250000	2.288750	4.183750	1.045250
max	1000.000000	2018.000000	40.863000	217.637000	14.375000

	Belgium	Brazil	Brunei Darussalam	Bulgaria	Cambodia
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
mean	1.839052	5.891926	0.246119	0.962194	0.076106
std	3.681559	10.635222	0.519793	4.016737	0.250004
min	0.001000	0.017000	0.001000	0.001000	0.001000
25%	0.175000	0.338750	0.004000	0.031000	0.004800
50%	0.573000	0.813500	0.013000	0.077000	0.013000
75%	1.786500	6.819500	0.188500	0.337750	0.049000
max	25.322000	76.500000	2.951000	34.399000	3.930000

KEY TRENDS AND OUTLIERS FROM THE EDA

Statistical Summary (mean, median, max, min)

- CO2 Emissions: U.S. max: 2512.9, Cambodia max: 3.93.
- Emissions Intensities: U.S. max: 9708.8, high values in Australia and Brazil.
- Emissions Multipliers: U.S. max: 10330.07, others much smaller.

EDA - Processes emissions data in multiple sheets

- Processing data for CO2 Emissions Intensities, Multipliers and Clustering
- Identify the top 5 emitting countries in each sheet (year).
- Provide insights into which countries contribute the most to emissions across the dataset.
- Data Summary

```
# Loop through all sheets
for sheet_name, df in cleaned_data.items():
    print(f"\nProcessing {sheet_name}...")

    # Select only numeric columns from column index 5 onward
    numeric_columns = df.iloc[:, 5:].select_dtypes(include=[np.number])

    # Check if there are numeric columns
    if numeric_columns.empty:
        print(f"No numeric data to process in {sheet_name}. Skipping...")
        continue

    # Aggregate emissions by country across industries
    country_totals = numeric_columns.sum(axis=0).sort_values(ascending=False)
    print(f"Top 5 emitting countries in {sheet_name}: \n{country_totals.head()}")
```

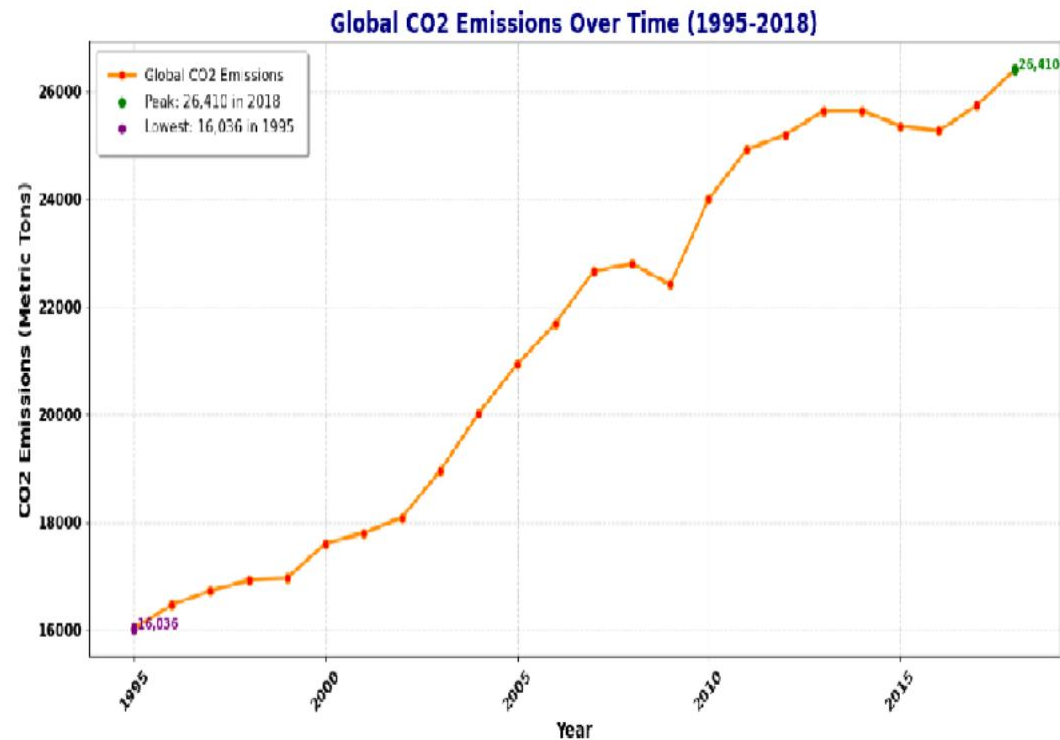
```
Processing CO2 Emissions...
Top 5 emitting countries in CO2 Emissions:
China, P.R.: Mainland    140074.3955
United States            103226.6670
Russian Federation       32733.9605
India                    29941.9350
Japan                    25063.5430
dtype: float64
```

```
Processing CO2 Emissions Intensities...
Top 5 emitting countries in CO2 Emissions Intensities:
Lao People's Dem. Rep.    3.379937e+06
Kazakhstan, Rep. of       3.180495e+06
Brunei Darussalam         2.417579e+06
Saudi Arabia              1.476459e+06
South Africa              1.321391e+06
dtype: float64
```

```
Processing CO2 Emissions Multipliers...
Top 5 emitting countries in CO2 Emissions Multipliers:
Kazakhstan, Rep. of       4.966566e+06
Russian Federation        2.646361e+06
South Africa              2.434185e+06
Saudi Arabia              2.111411e+06
China, P.R.: Mainland     2.011595e+06
dtype: float64
```

```
Processing Clustering...
Top 5 emitting countries in Clustering:
F1998    6.143740e+06
F1995    6.065450e+06
F1999    6.024472e+06
F1996    5.956289e+06
F1997    5.724594e+06
dtype: float64
```

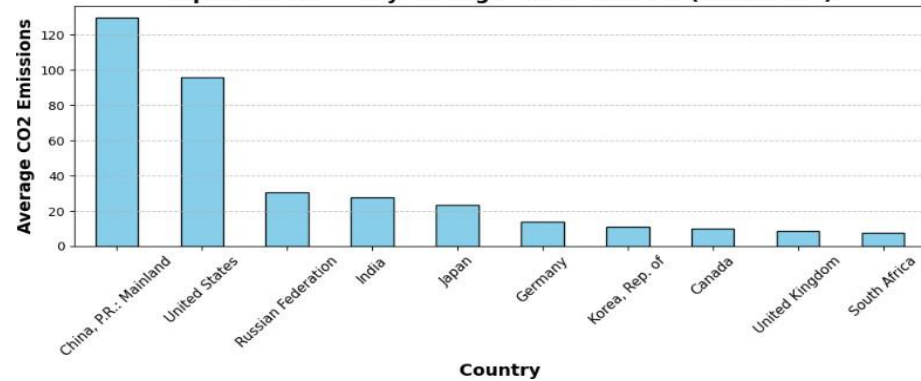

Global CO2 Emissions Over Time



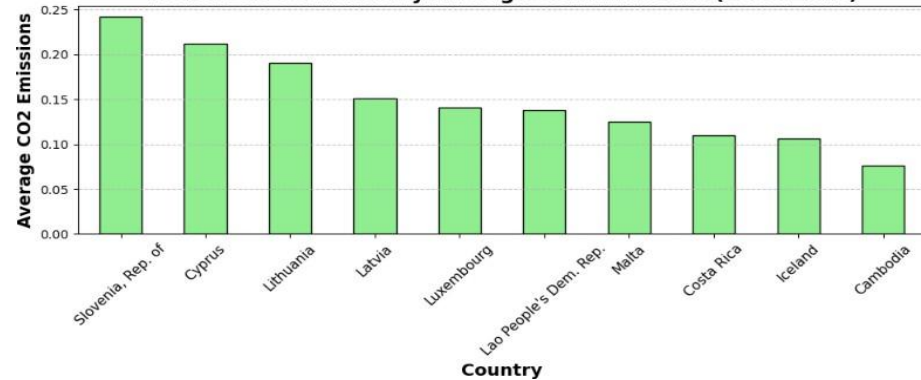
- **Rising Trend from 1995 to 2018**
- **Peak Emissions – Green mark**
- **Lowest Emissions – Purple mark**
- **Emission Dynamics**

Data Visualization - Country-Level Emissions Analysis

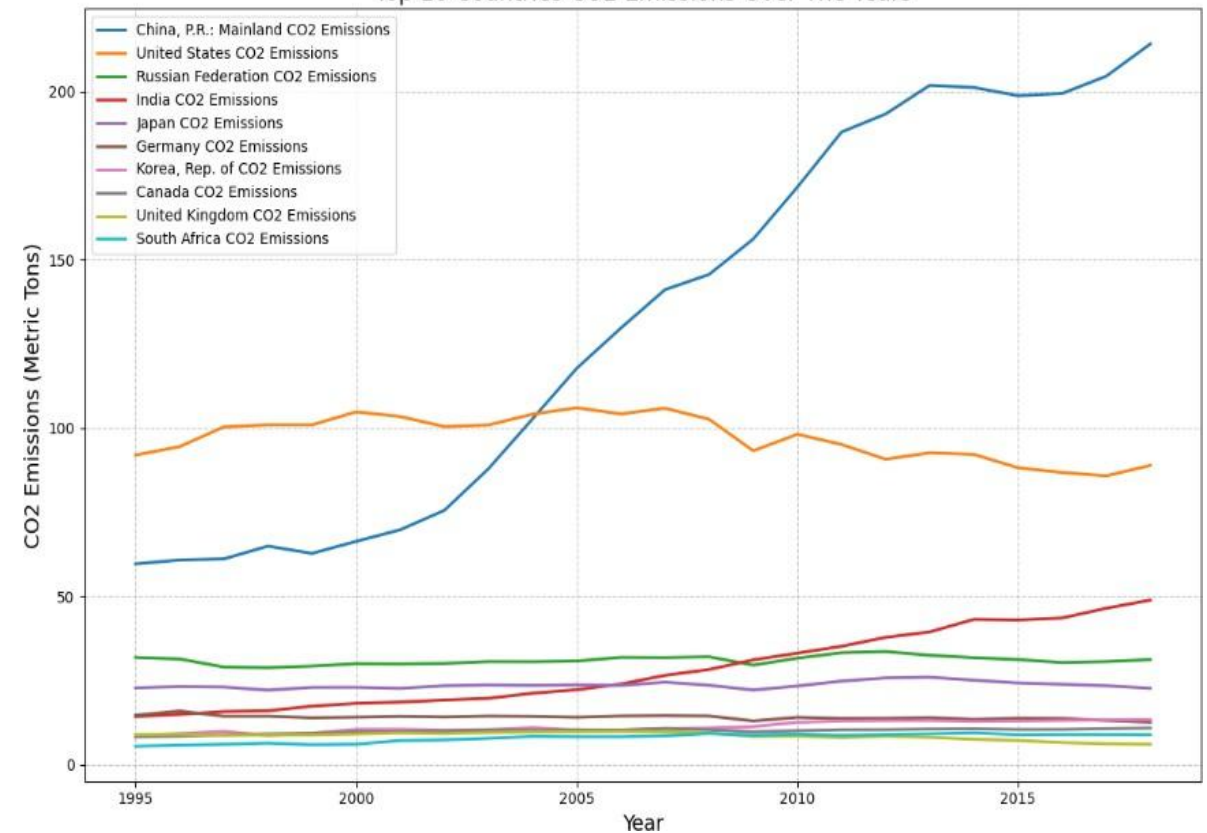
Top 10 Countries by Average CO2 Emissions (1995-2018)



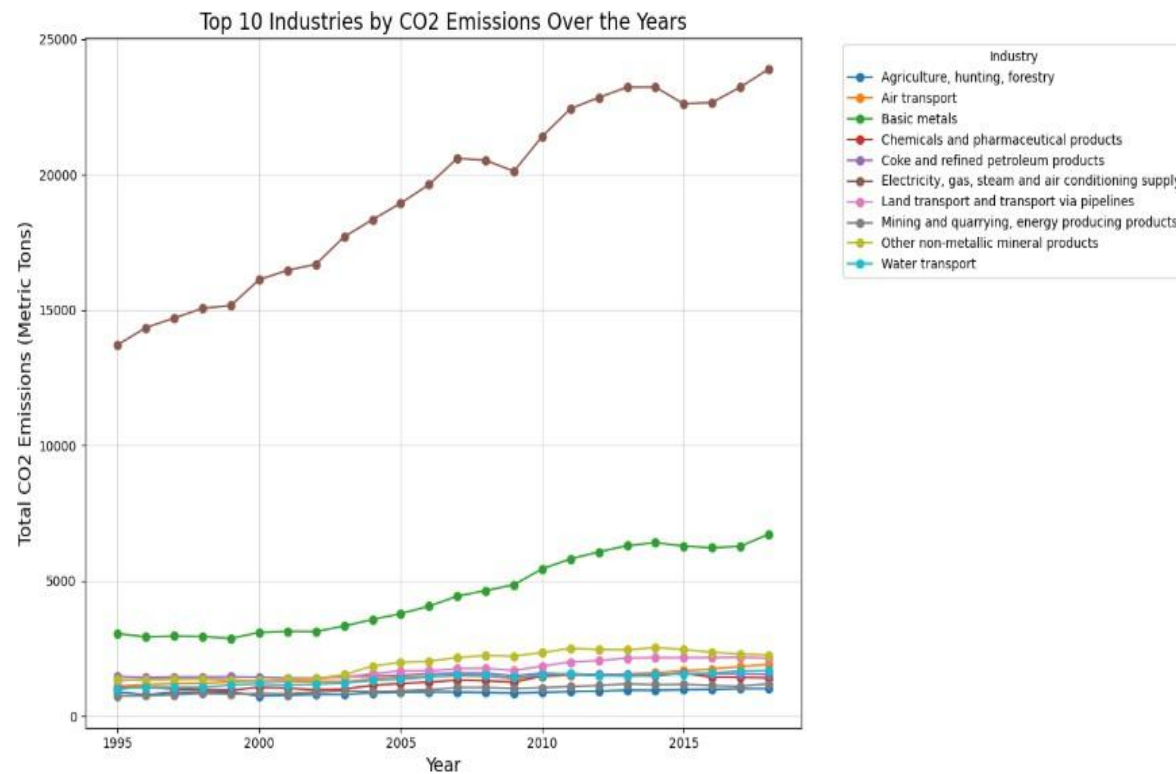
Bottom 10 Countries by Average CO2 Emissions (1995-2018)



Top 10 Countries CO2 Emissions Over The Years



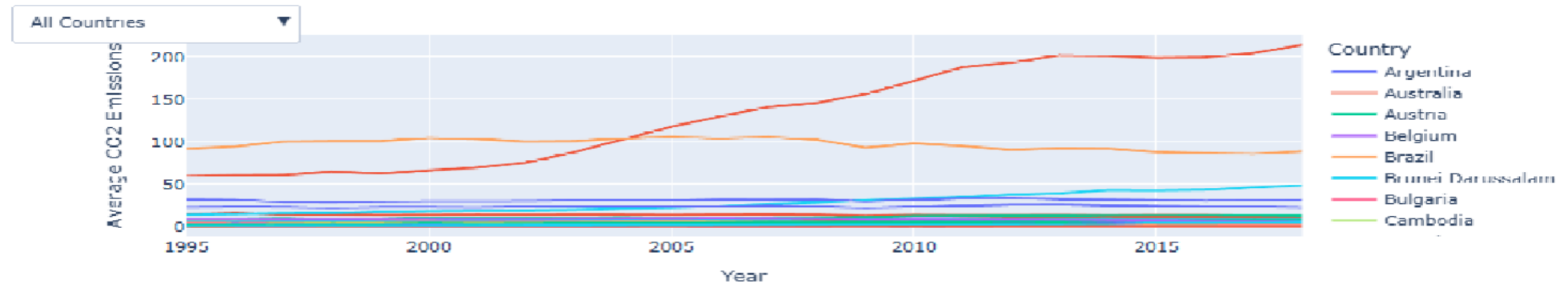
Data Visualization - Industry-Level Emissions Analysis



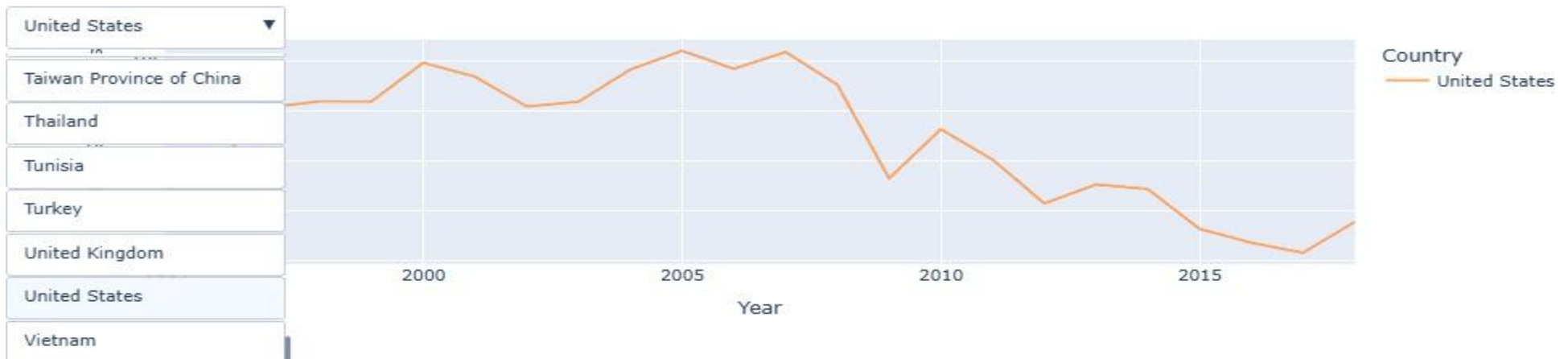
- **Aggregated CO2 emissions across all industries over the years**
- **Highest CO2 emissions impact**
- **Largest contributors to CO2 emissions (e.g., Energy, Manufacturing)**
- **Global Impact**

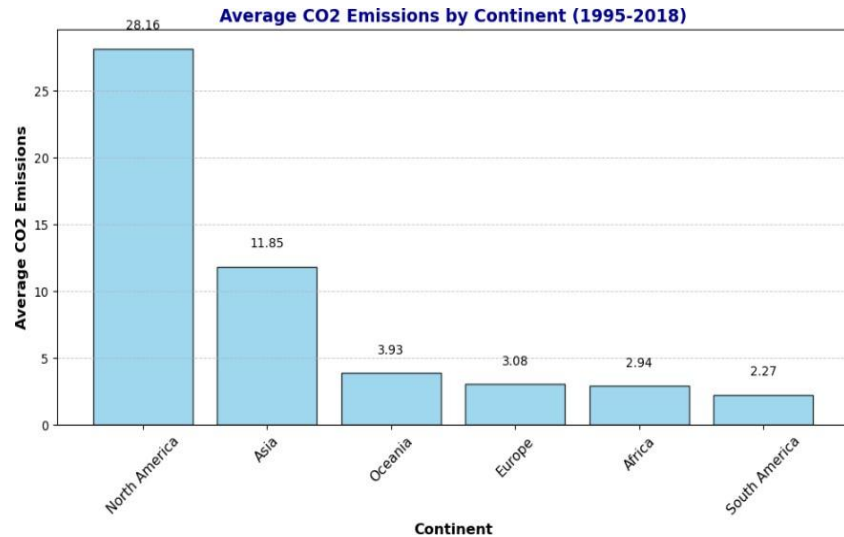
Average CO2 Emissions for all countries

Average CO2 Emissions Over Time (1995-2018)



Average CO2 Emissions Over Time (1995-2018)





Average CO2 Emissions by Continent (1995-2018):

Continent

North America 28.157711

Asia 11.848396

Oceania 3.926867

Europe 3.083828

Africa 2.936610

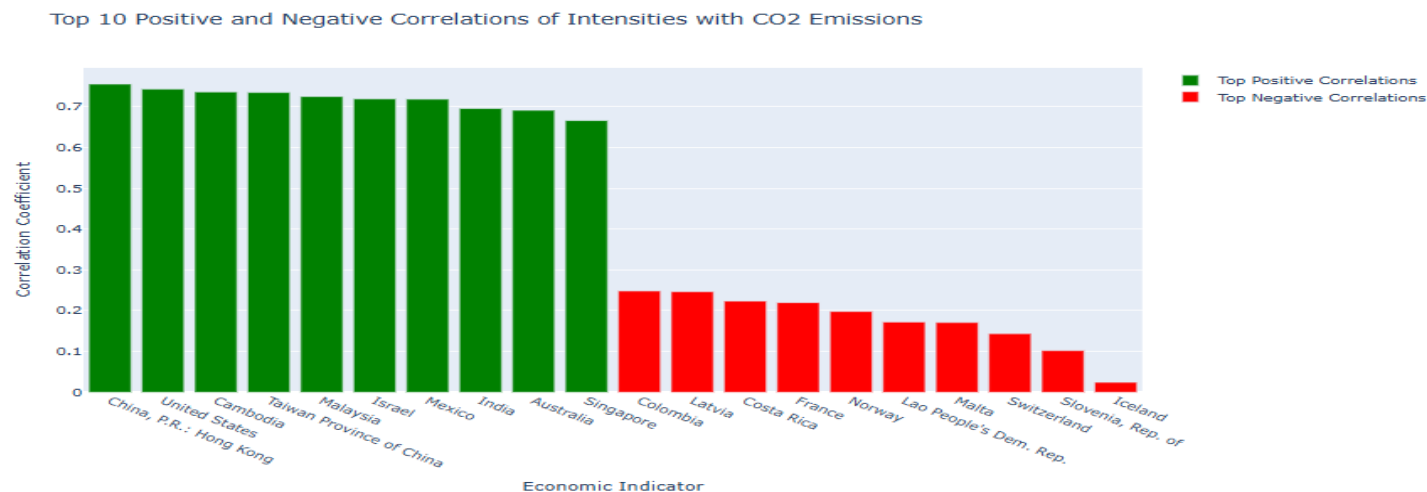
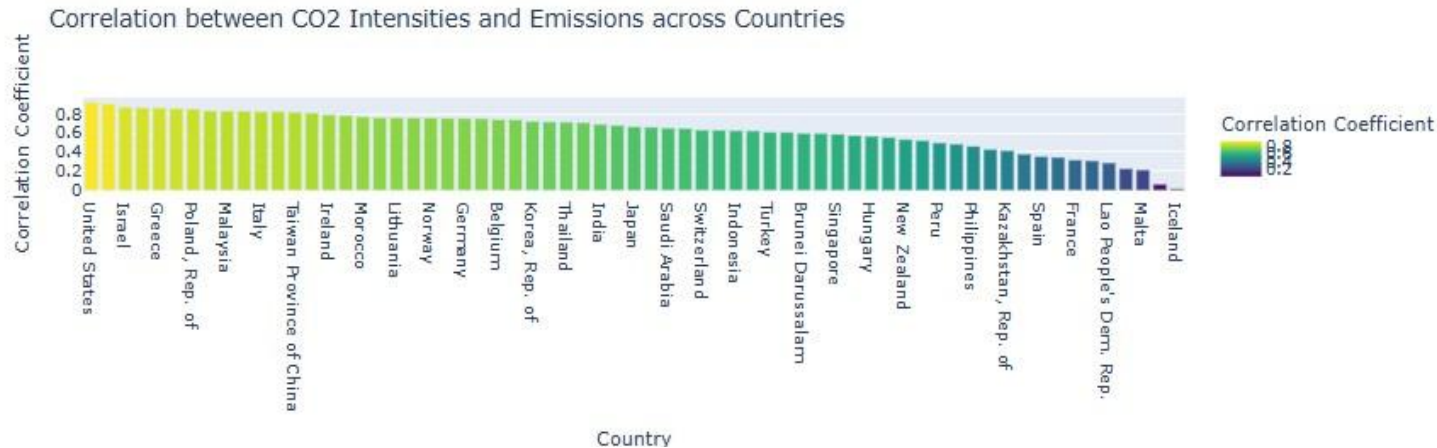
South America 2.271485

Name: Emissions, dtype: float64

Data Visualization - Emissions Analysis by Continent

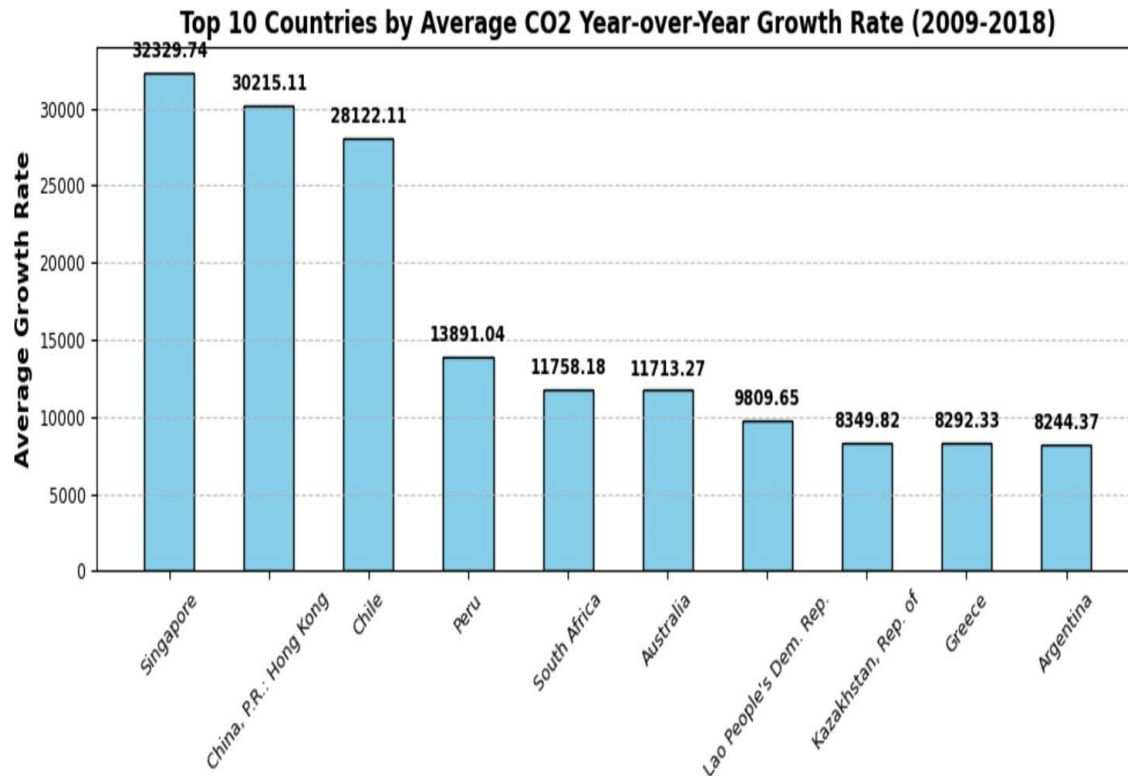
- Comparison between emissions across regions specifically continents by their average CO2 Emissions.
- **Highest emissions** - North America
- **Higher emissions** - Asia
- **Moderate emissions** – Oceania
- **Lower emissions** – Europe
- **Lowest emissions** - Africa & South America

Correlation Analysis - Data Correlation



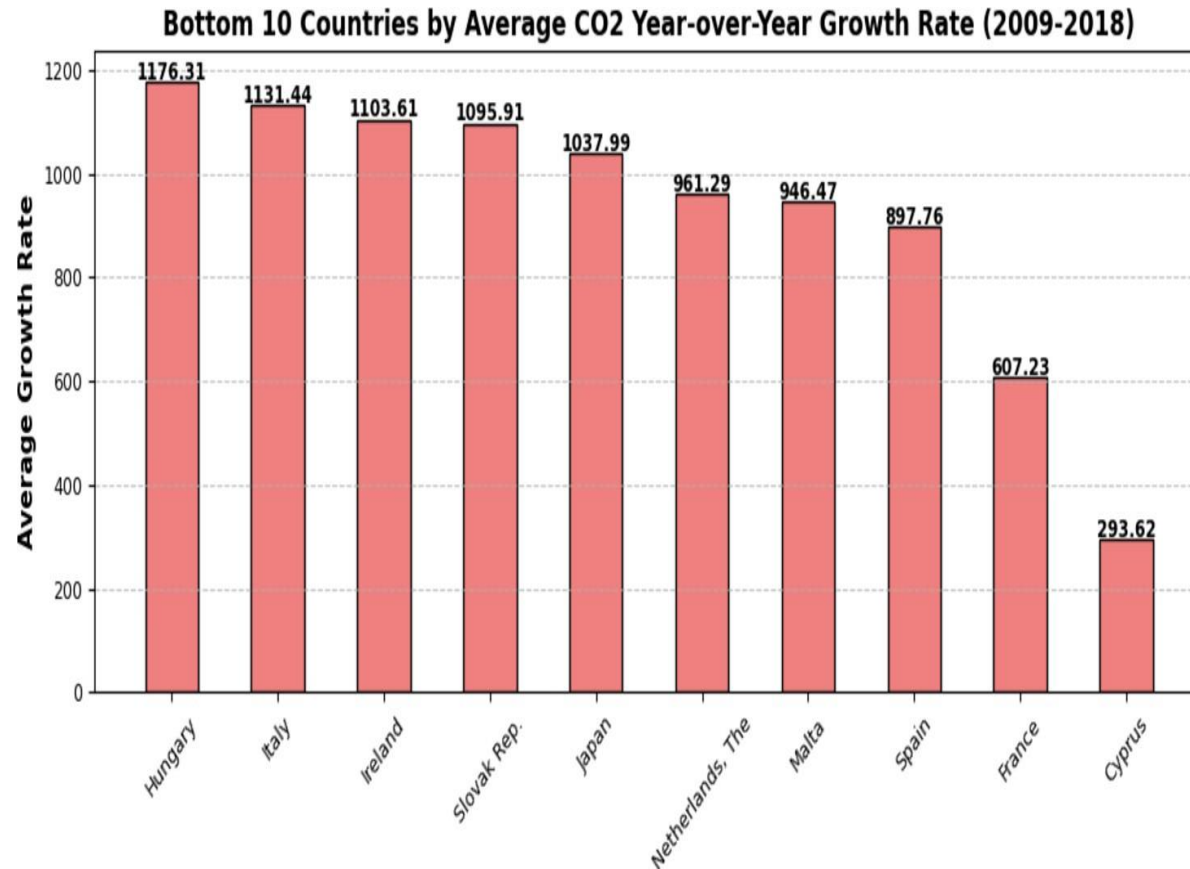
- **Strong Positive Correlations (Above 0.7) - Green Gradient**
- **Moderate Correlations (0.4 - 0.7) - Yellow Gradient**
- **Weaker Correlations (Below 0.4) - Blue Gradient**
- **Economic Growth Correlated with Rising CO2 Levels**

Feature Engineering - Top 10 Countries by Average CO2 Year-over-Year Growth (2009–2018)



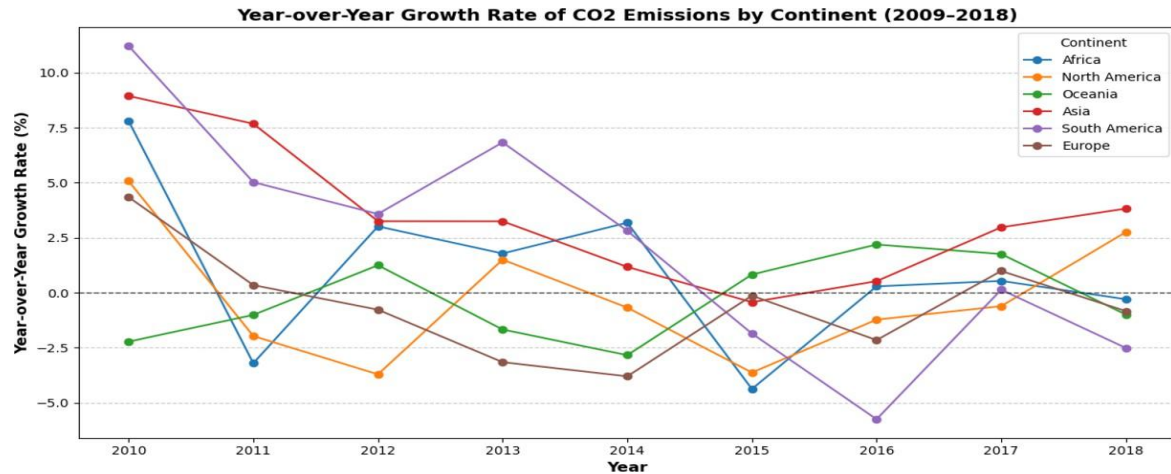
- **Singapore leads the pack:**
- Singapore exhibits the highest average CO2 growth rate at **32,329.74%**, signaling rapid industrialization and urbanization.
- **Asia dominates the top ranks:**
- Both **Hong Kong (30,215.11%)** and **Lao People's Democratic Republic (9,809.65%)** reinforce Asia's growing share in global emissions.
- **Emerging economies show high growth:**
- Countries like **Chile (28,122.11%)**, **Peru (13,891.04%)**, and **Kazakhstan (8,349.82%)** highlight South America and Central Asia's increasing contributions.

Feature Engineering- Bottom 10 Countries by Average CO2 Year-over-Year Growth (2009–2018)

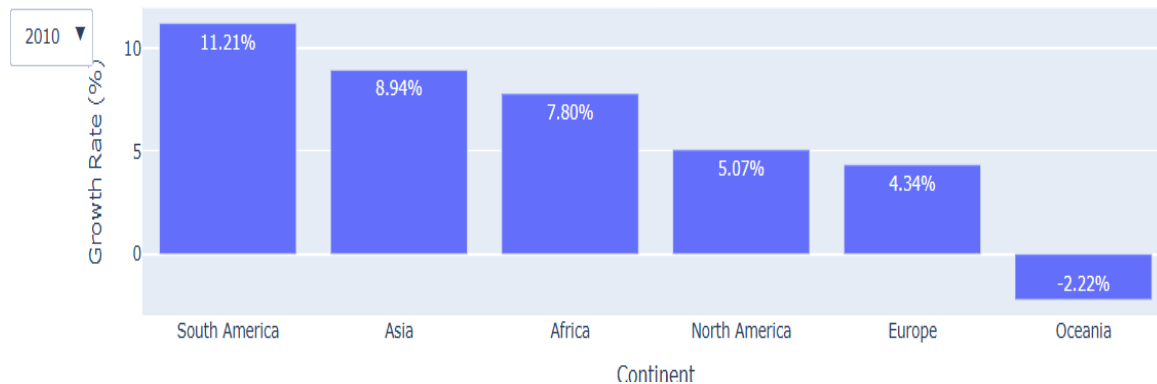


- **Hungary leads the slow growth group:**
- Hungary tops the bottom 10 with an average growth rate of **1,176.31%**, the highest among the slow-growing nations.
- **Western Europe dominates the bottom 10:**
- Countries like **Italy (1,131.44%)**, **Ireland (1,103.61%)**, and **France (607.23%)** indicate a slower increase in CO2 emissions compared to emerging economies.
- **Stabilized emissions in developed nations:**
- Japan (1,037.99%) and the Netherlands (961.29%) demonstrate controlled growth, reflecting their efforts in energy efficiency and renewable energy adoption.
- **Cyprus shows the least growth:**
- Cyprus has the lowest average growth rate among all countries at **293.62%**, signaling minimal increases in emissions over the decade

CO2 Growth rate by continent

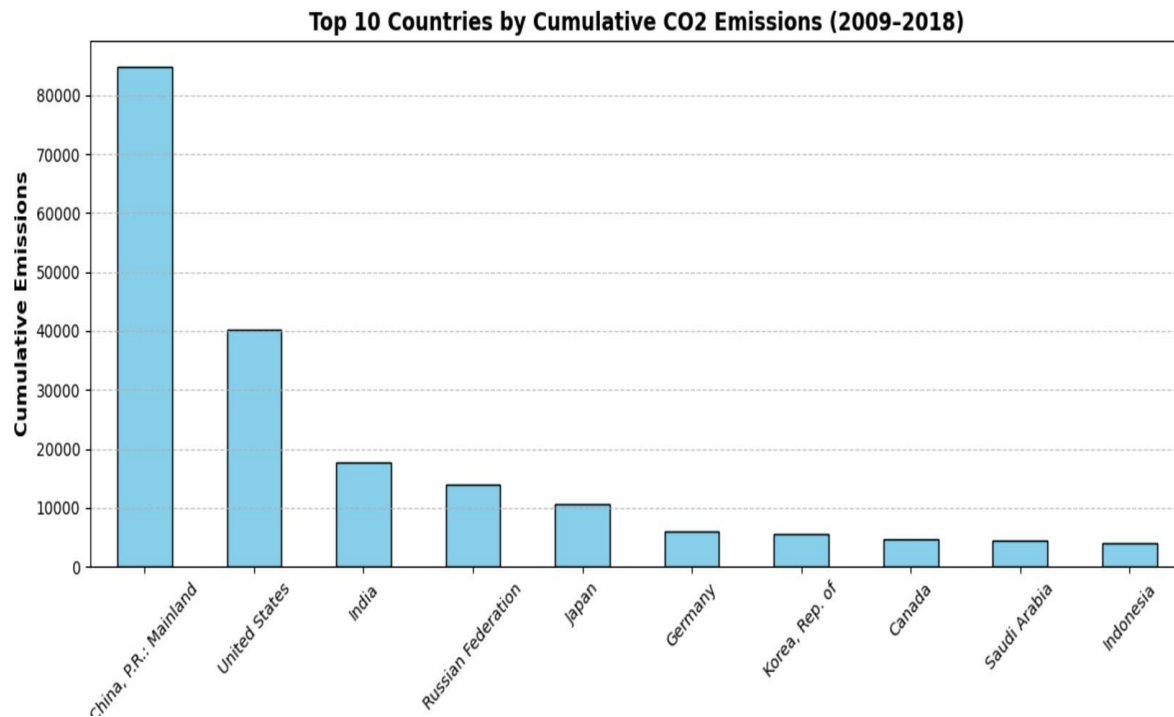


Year-over-Year CO2 Emissions Growth Rate in 2010



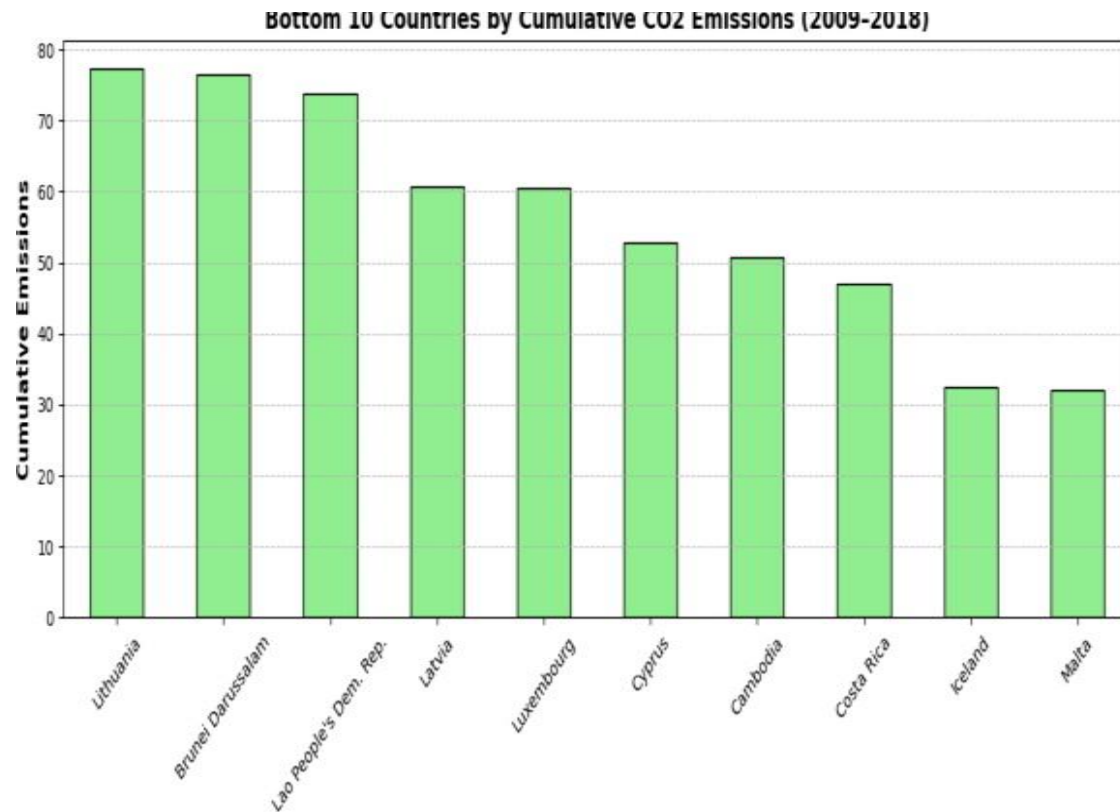
- **South America leads in 2010:**
- South America recorded the highest growth rate in 2010 at **11.21%**, highlighting rapid industrial and agricultural expansion.
- **Asia's consistent growth:**
- Asia maintained a steady upward trend throughout the period, driven by industrialization and population growth, cementing its position as a global emissions leader.
- **Declining growth in Oceania and Europe:**
- Oceania and Europe show **negative or minimal growth rates**, reflecting the impact of stringent environmental policies and cleaner energy transitions.
- **Africa shows potential growth:**
- Africa's emissions growth fluctuates, peaking at **7.8% in 2010**, indicating developing economies with increasing energy demands.

Feature Engineering - Top 10 Countries by Cumulative CO2 Emissions (2009–2018)



- **China leads by a significant margin:**
- China is the largest contributor with **over 80,000 metric tons** of cumulative CO2 emissions, reflecting its vast industrial base and population size.
- **The United States and India follow:**
- The United States ranks second, contributing roughly **half of China's total emissions**, while India secures the third spot due to rapid industrial growth.
- **Other major contributors:**
- Countries like **Russia, Japan, and Germany** make up the mid-tier of top emitters, showcasing their large-scale industrial activities.
- **Developed vs. developing nations:**
- The chart shows a mix of developed (e.g., United States, Japan) and developing (e.g., India, Indonesia) nations, highlighting the global spread of emissions responsibility.

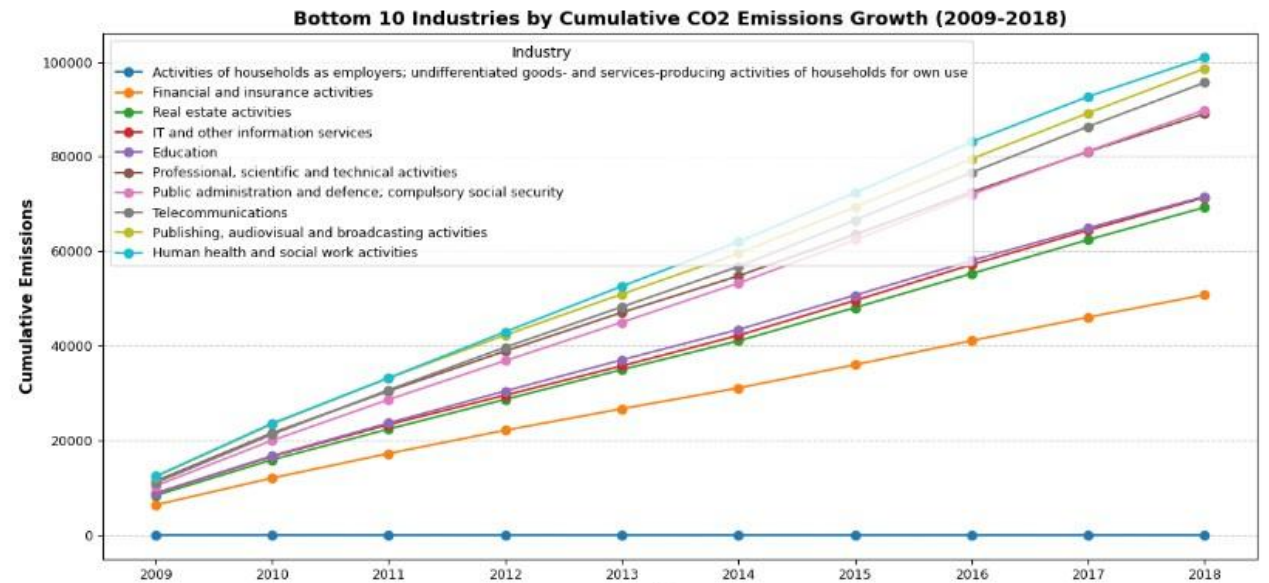
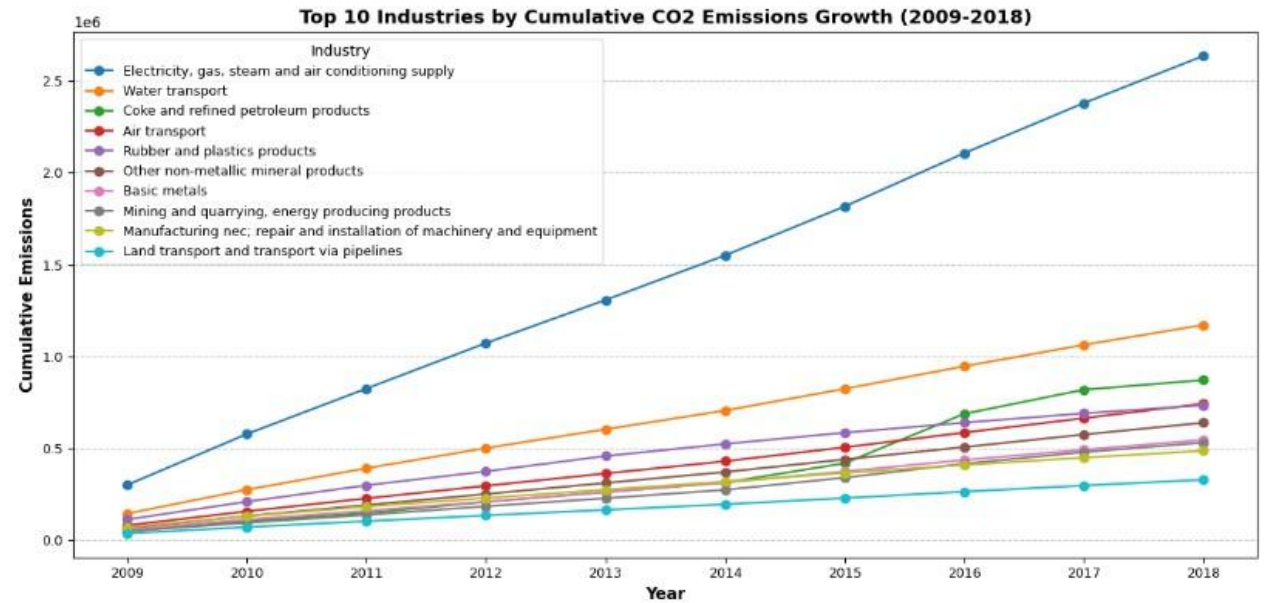
Feature Engineering- Bottom 10 Countries by Cumulative CO2 Emissions (2009–2018)



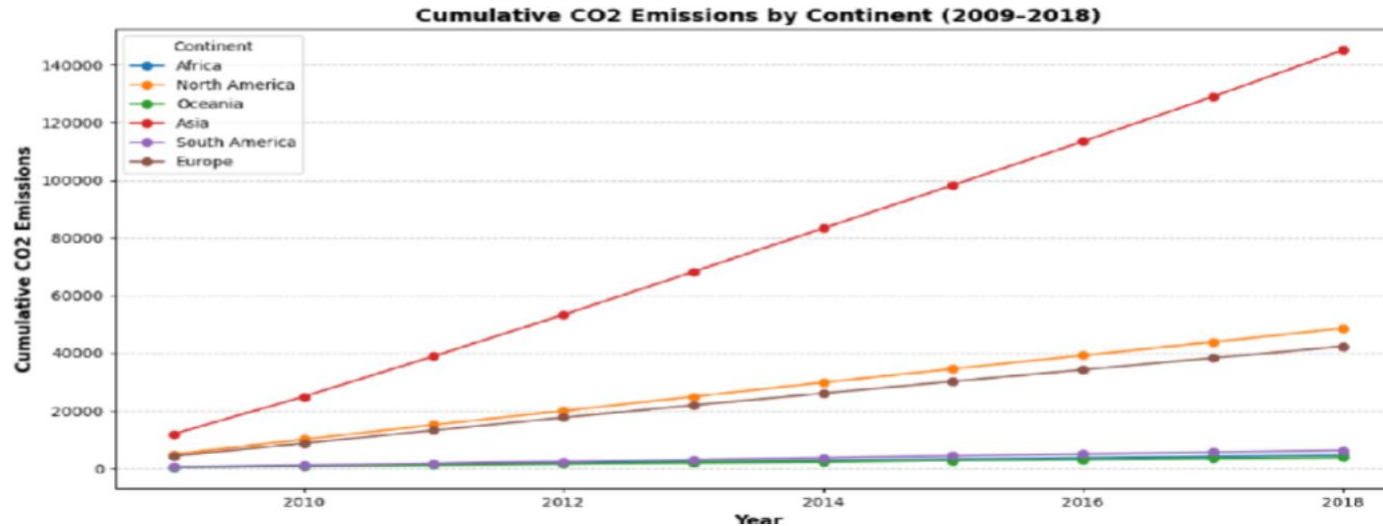
- **Low emissions reflect smaller economies:**
- Countries like **Lithuania (top of the list)** and **Malta (bottom)** have limited industrial activities, resulting in low cumulative CO2 emissions.
- **Dominance of small nations:**
- Most countries on this list are smaller nations, such as **Brunei Darussalam, Luxembourg, and Cyprus**, with lower energy demands and less industrial infrastructure.
- **Minimal global contribution:**
- Collectively, these countries contribute only a fraction of global emissions compared to major emitters, emphasizing the disparity in CO2 emissions responsibility.
- **Role of renewable energy adoption:**
- Some countries, like **Iceland**, benefit from the use of renewable energy sources such as geothermal and hydropower, further reducing their emissions.

Top & Bottom 10 Industries by Cumulative CO2 Emissions Growth

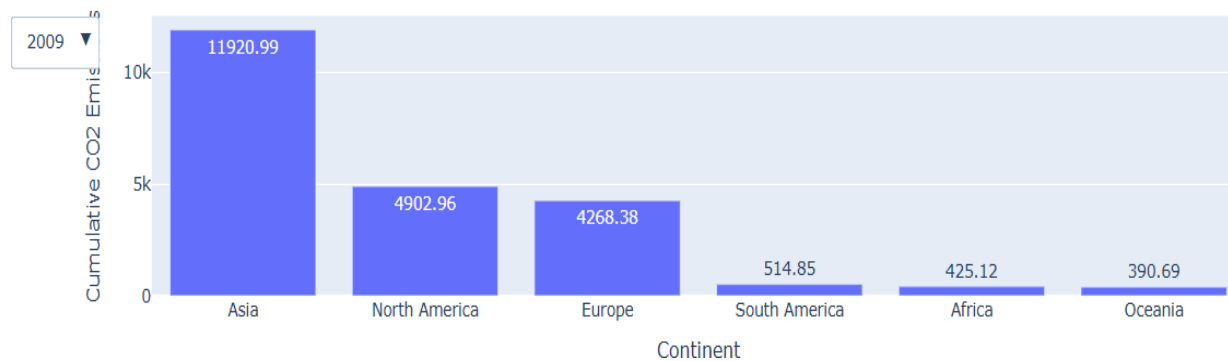
- **Energy and transport sectors dominate top emitters:**
- Industries like **electricity, gas, and air conditioning**, **water transport**, and **air transport** drive the highest emissions, reflecting their heavy reliance on fossil fuels.
- **Service-based and knowledge industries have minimal impact:**
- Bottom industries like **education**, **telecommunications**, and **financial services** exhibit the lowest emissions due to low energy requirements.



Cumulative Emissions by Continent



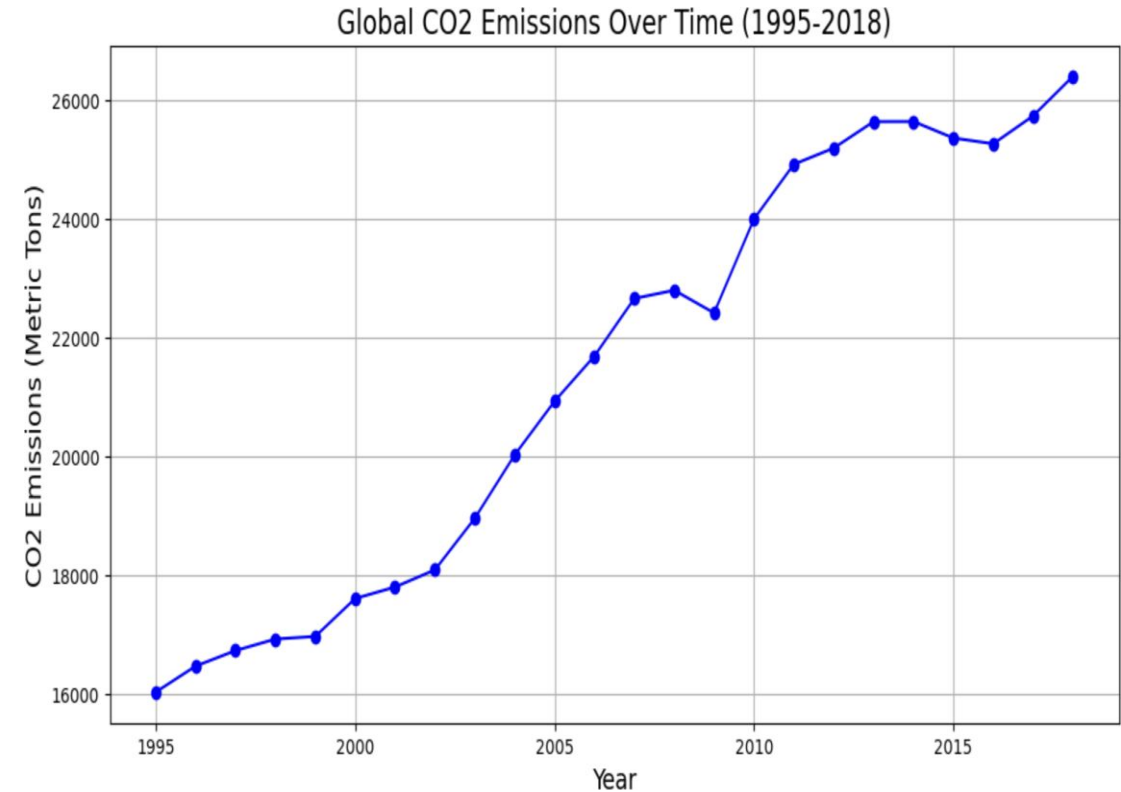
Cumulative CO2 Emissions in 2009



- **Asia dominates emissions globally:**
- Asia exhibits the highest cumulative CO2 emissions, with a consistent upward trend, far outpacing other continents by 2018.
- **North America and Europe rank next:**
- North America and Europe follow as major contributors, though their growth rates are slower compared to Asia.
- **Lower emissions in other continents:**
- Africa, Oceania, and South America have significantly lower emissions, reflecting smaller industrial bases and lower energy consumption.

Predictive Modeling

- **Aggregate global emissions**
- **Group by year and sum emissions**
- **Plot global emissions over time**
- **Stationary or non – stationary?**



ADF Statistic: -3.118086477356265
p-value: 0.025243341036345485
The time series is stationary.

ARIMA

- Training set
- The model uses:
 - 1 lag for the autoregressive (AR) part.
 - First-order differencing ($d=1$) to make the series stationary.
 - 1 lag for the moving average (MA) part.
- Log Likelihood: -166.165
- AIC (338.331) and BIC (341.318):

```
from statsmodels.tsa.arima.model import ARIMA

# Split the data into train and test sets
train = global_emissions[global_emissions['Year'] <= 2015]
test = global_emissions[global_emissions['Year'] > 2015]

# Build the ARIMA model (parameters p, d, q need to be determined)
model = ARIMA(train['Global Emissions'], order=(1,1,1))
model_fit = model.fit()
print(model_fit.summary())
```

SARIMAX Results

Dep. Variable:	Global Emissions	No. Observations:	21
Model:	ARIMA(1, 1, 1)	Log Likelihood	-166.165
Date:	Thu, 28 Nov 2024	AIC	338.331
Time:	17:56:43	BIC	341.318
Sample:	0	HQIC	338.914
	- 21		
Covariance Type:	opg		

	coef	std err	z	P> z	[0.025	0.975]
ar.L1	0.9957	0.058	17.229	0.000	0.882	1.109
ma.L1	-0.9767	0.173	-5.661	0.000	-1.315	-0.639
sigma2	9.564e+05	1.59e-08	6.02e+13	0.000	9.56e+05	9.56e+05

Ljung-Box (L1) (Q):	0.64	Jarque-Bera (JB):	0.40
Prob(Q):	0.42	Prob(JB):	0.82
Heteroskedasticity (H):	8.02	Skew:	0.30
Prob(H) (two-sided):	0.01	Kurtosis:	2.64

Forecasting 3 years (2016-2018)

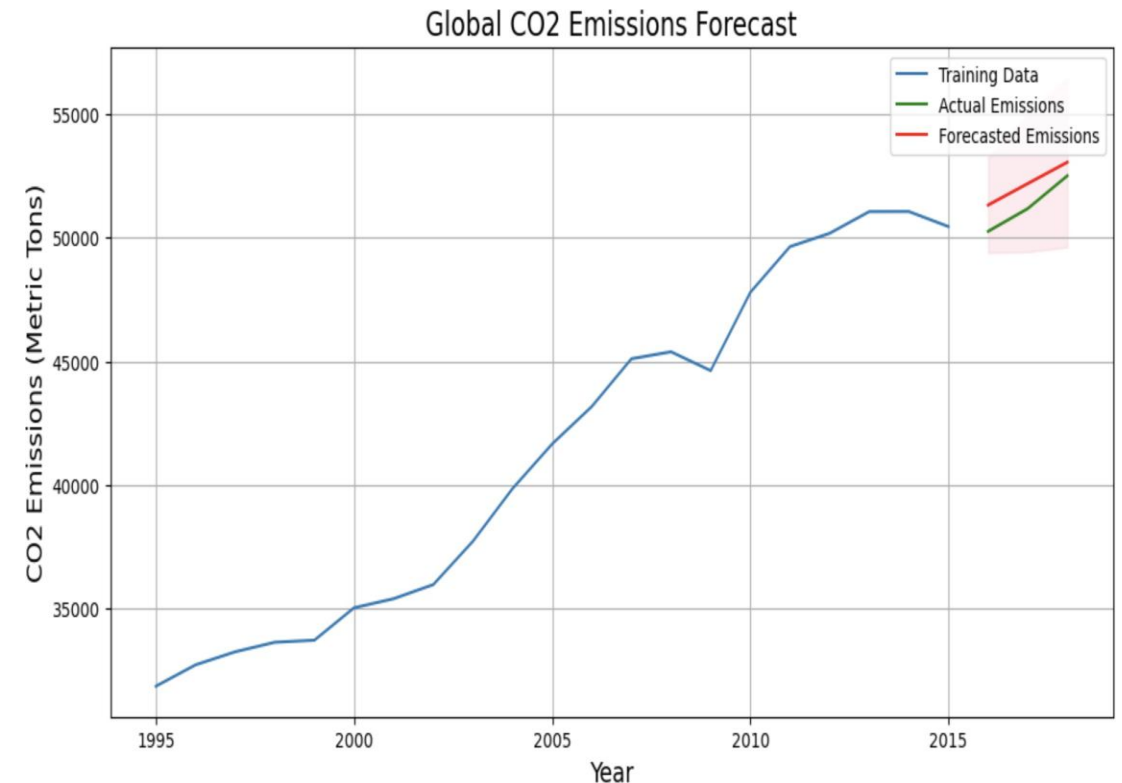
- Blue Line (Training Data)
- Green Line (Actual Emissions)
- Red Line (Forecasted Emissions)
- Shaded Area (Confidence Interval)
- MAE (875.30 metric tons)
- MSE (820,358.03 metric tons²)

```
from sklearn.metrics import mean_squared_error, mean_absolute_error

# Calculate evaluation metrics
mse = mean_squared_error(test['Global Emissions'], forecast.predicted_mean)
mae = mean_absolute_error(test['Global Emissions'], forecast.predicted_mean)
print(f'Mean Squared Error: {mse:.2f}')
print(f'Mean Absolute Error: {mae:.2f}')
```

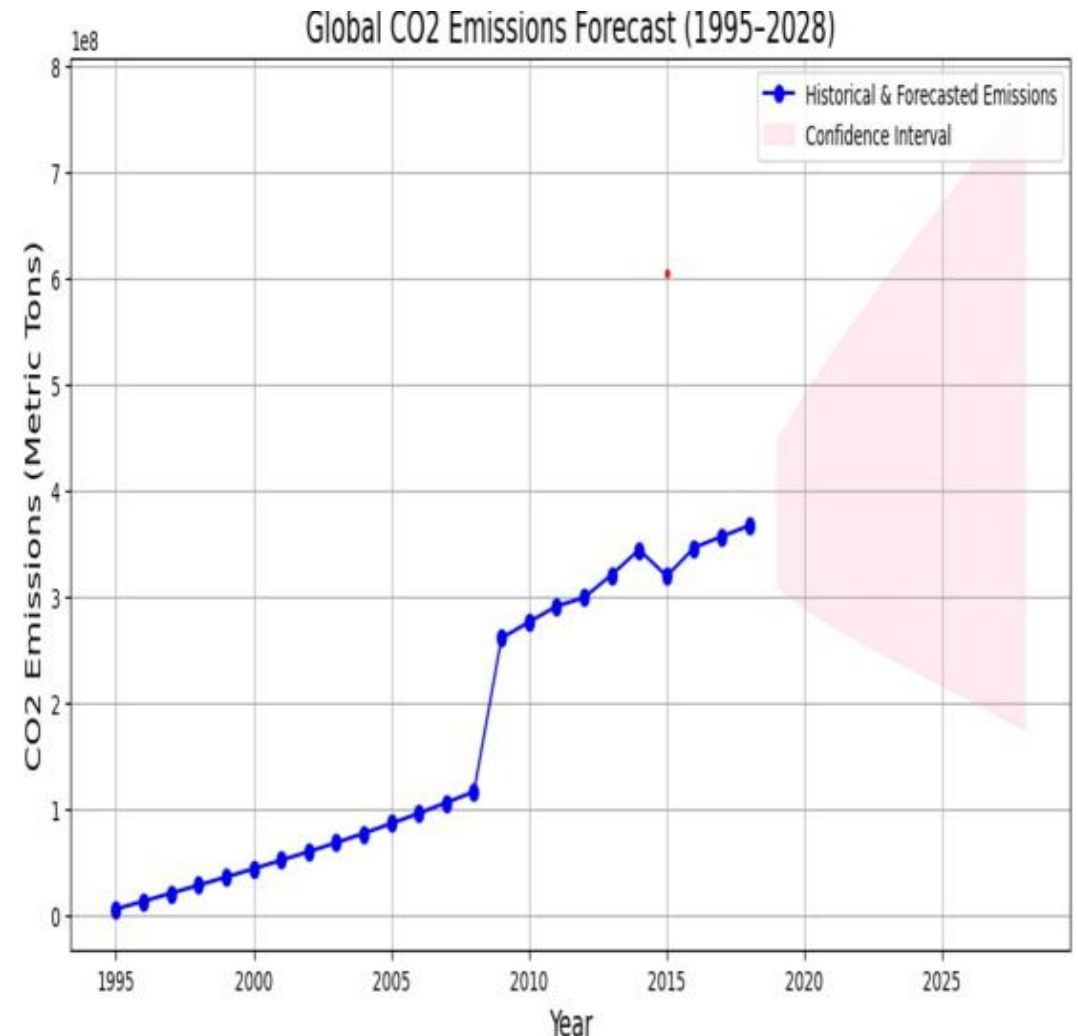
Mean Squared Error: 820358.03

Mean Absolute Error: 875.30

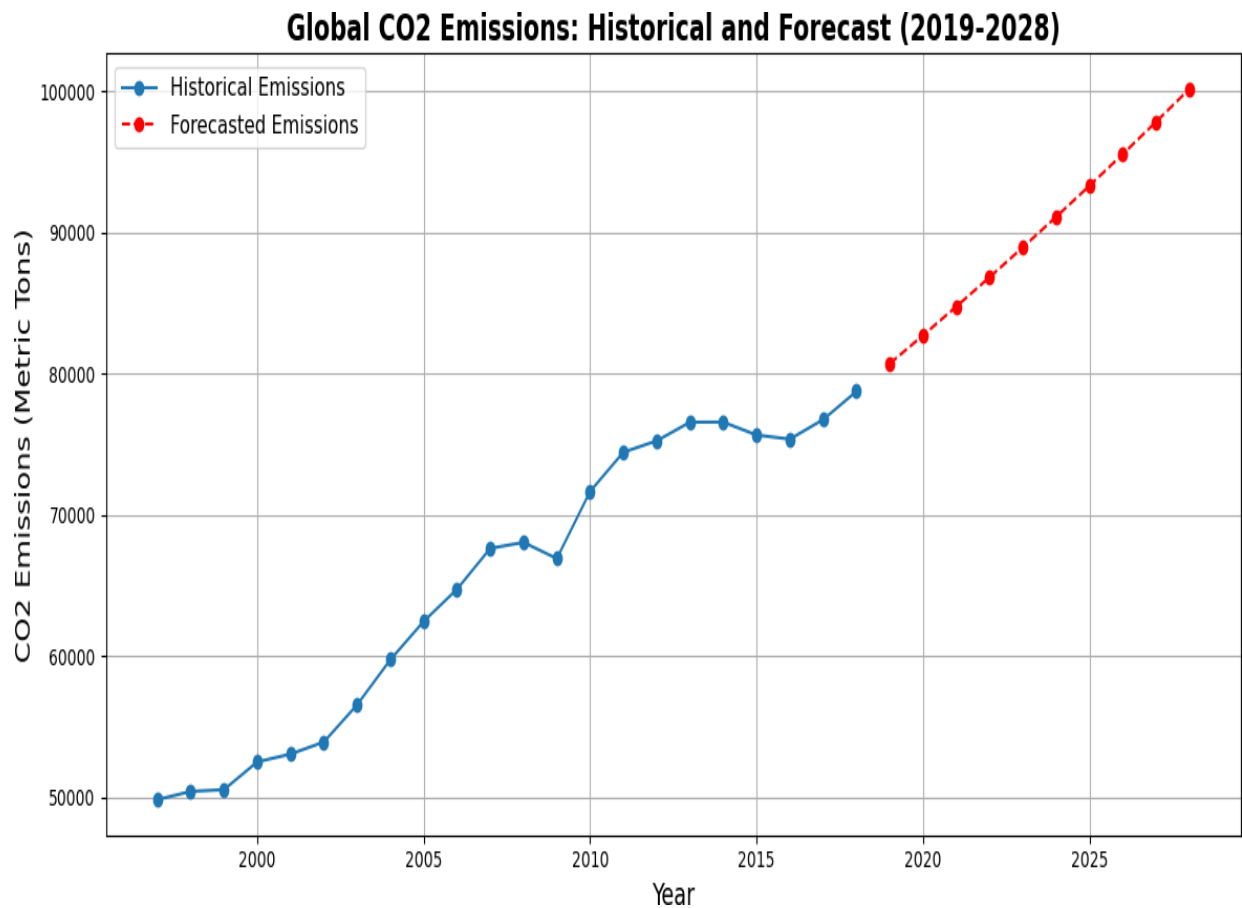


Predicting for the next decade (2019-2028)

- Historical trend shows steady growth:
- From 1995 to 2018, global CO2 emissions steadily increased, reflecting industrialization and energy demands worldwide.
- Projected rise through 2028:
- Forecasted emissions show a continued upward trajectory, with a wide confidence interval highlighting potential variability based on global policies and technological advancements.
- Uncertainty in future trends:
- The confidence interval suggests emissions could reach significantly higher levels, emphasizing the critical need for proactive mitigation measures.



Forecasting using Regression Model 2019 - 2028



Year	Forecasted Emissions
2019	80,735.54
2020	82,738.06
2021	84,778.35
2022	86,857.24
2023	88,975.48
2024	91,133.81
2025	93,332.99
2026	95,573.79
2027	97,856.99
2028	100,183.41

Predictive Techniques and Forecasting Models

- **SARIMAX**
- **ARIMA (Auto-Regressive Integrated Moving Average):** Country-specific CO2 emissions forecast
- **Clustering (K-Means):** Grouping countries based on similar CO2 emission patterns
- **EDA (Exploratory Data Analysis) with Growth Rate Computation:** EDA involved: Computing year-over-year growth rates for CO2 emissions and Identifying trends and anomalies in emissions data
- **Correlation Analysis:** Analyzed the correlation between indicators and CO2 emissions to identify predictors and potential relationships
- Statistical Techniques like: Stationarity Tests (**ADF Test**): Verified that the time series was suitable for ARIMA modeling and **AIC/BIC** Optimization-Used to select the best-fitting ARIMA model

Identifying Clusters of Countries Based on CO2 Emission Trends



Feature Engineering



Computed Average Emissions and Emission Variance for each country



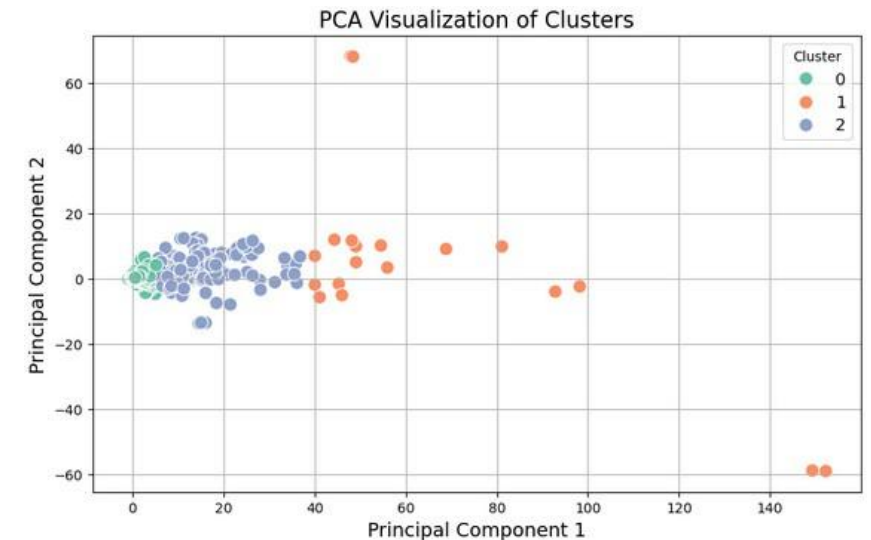
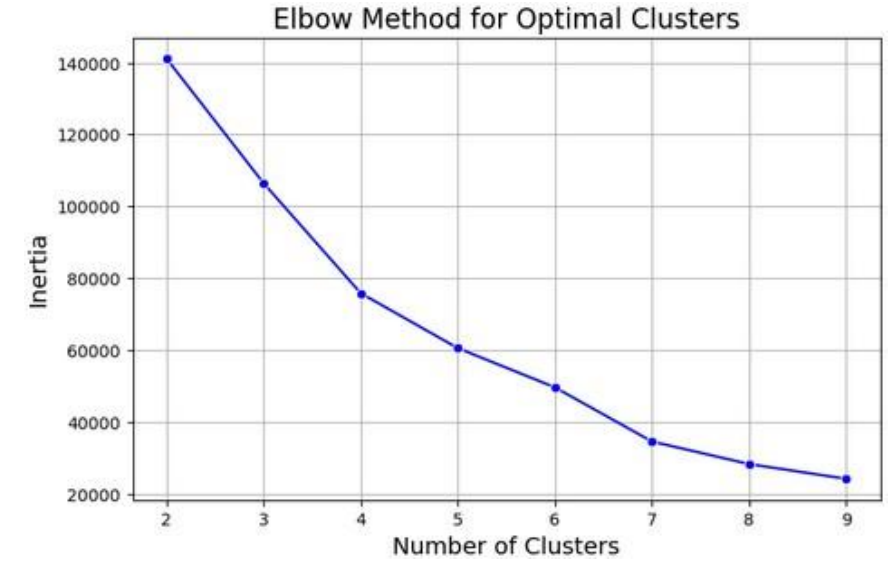
Clustering Approach:



Used K-Means with an optimal cluster count determined by the Elbow Method



Principal Component Analysis (PCA) for 2D visualization of clusters

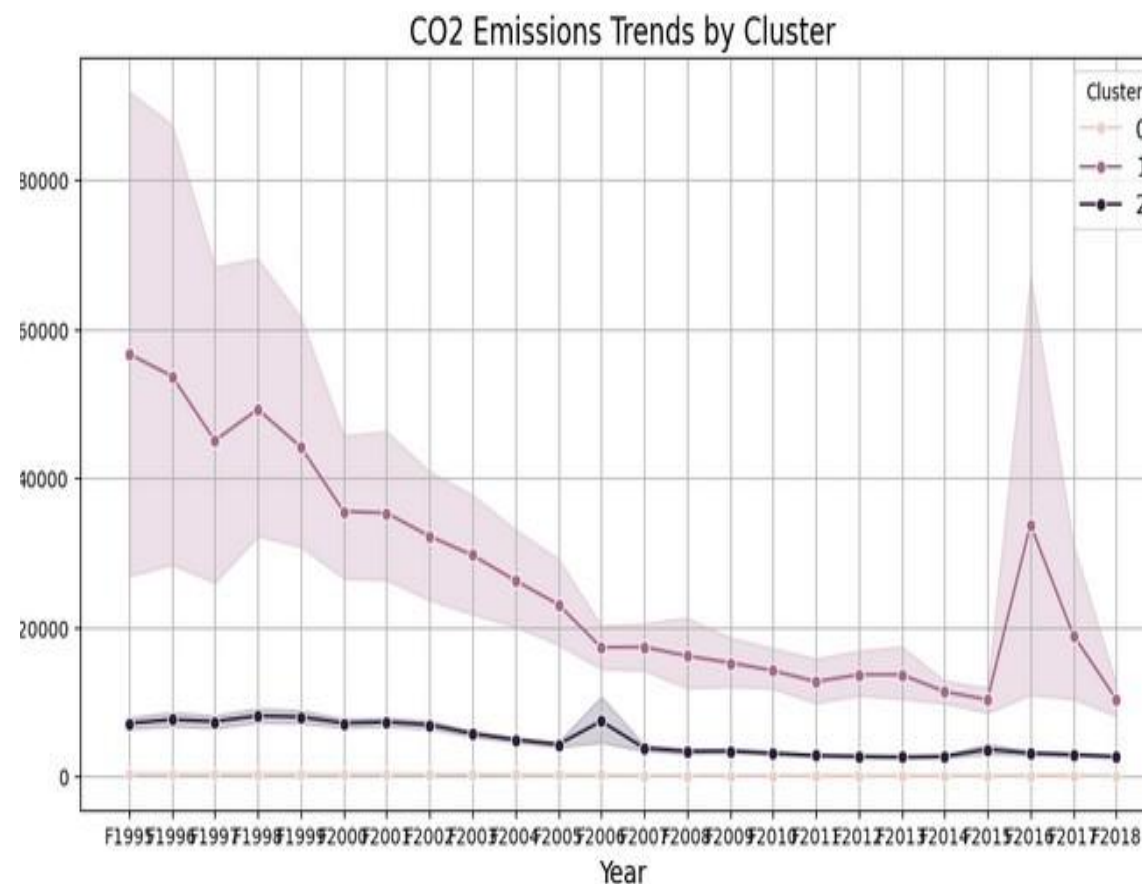


Results of Clustering

- **Cluster 0:** Low emissions with minimal variance
- **Cluster 1:** High emissions with significant variance and global policy impact
- **Cluster 2:** Moderate emissions with medium variance
- PCA Visualization provides a clear separation of clusters

Cluster Summary:

	Average_Emissions	Emission_Variance
Cluster		
0	221.129784	9.698265e+04
1	26488.213000	9.279823e+08
2	4945.485570	4.476165e+07



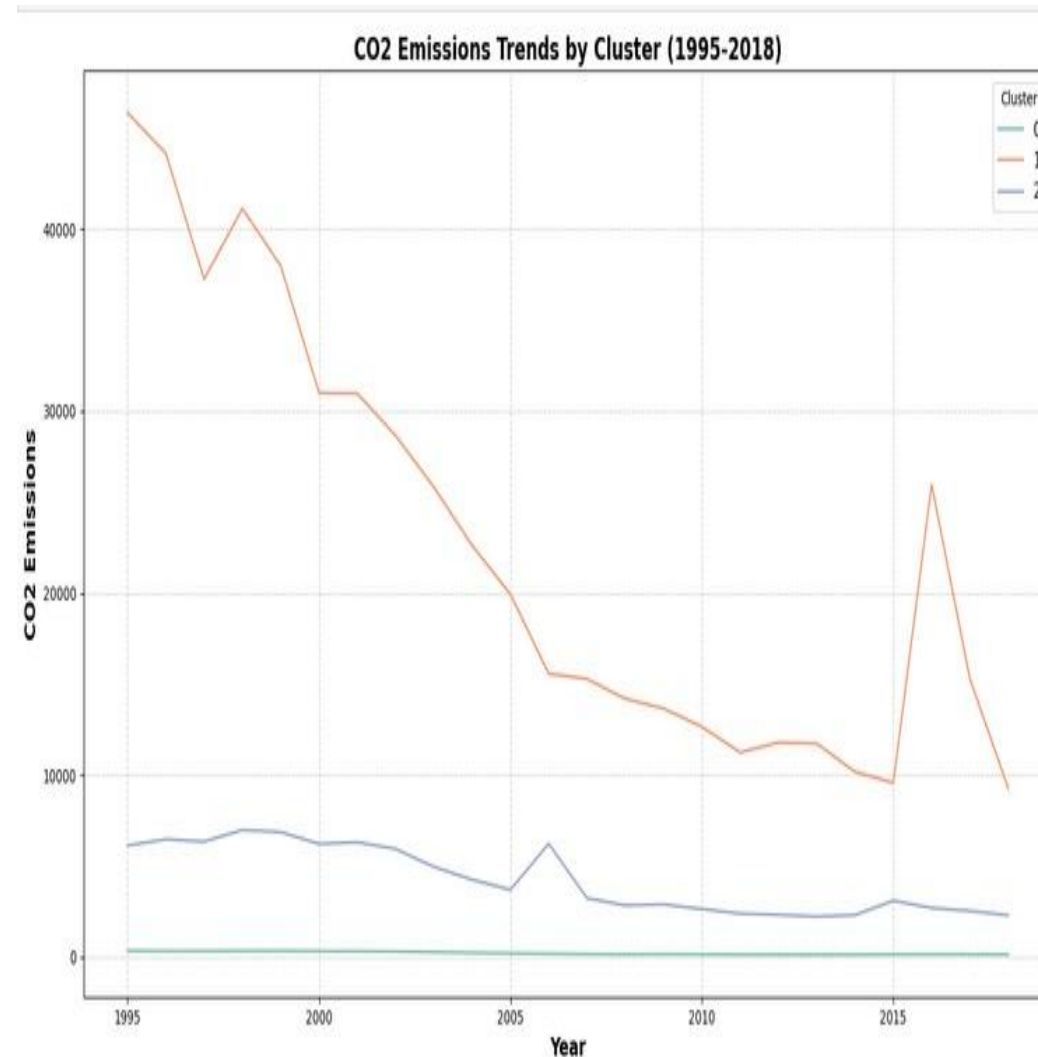
Cluster Trends & Correlation

CO2 Emission Trends:

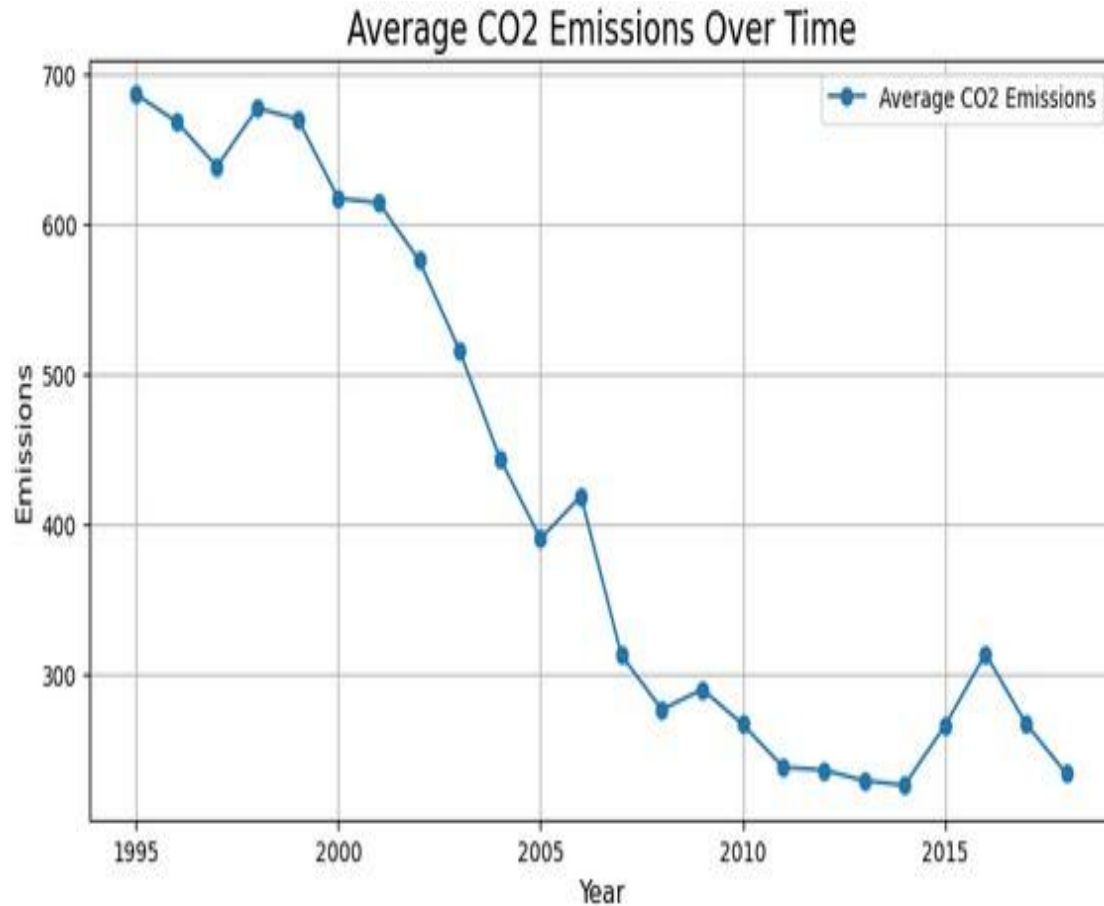
- Cluster-wise analysis (1995–2018) to visualize emission reductions and policy impacts

Key Insights:

- Cluster 0: Countries with stabilized emissions, possibly result of stringent policies
- Cluster 2: Gradual reductions indicating evolving climate policies
- Cluster 1: High emitters with recent decline due to global agreements



Forecasting CO2 Emissions Using ARIMA



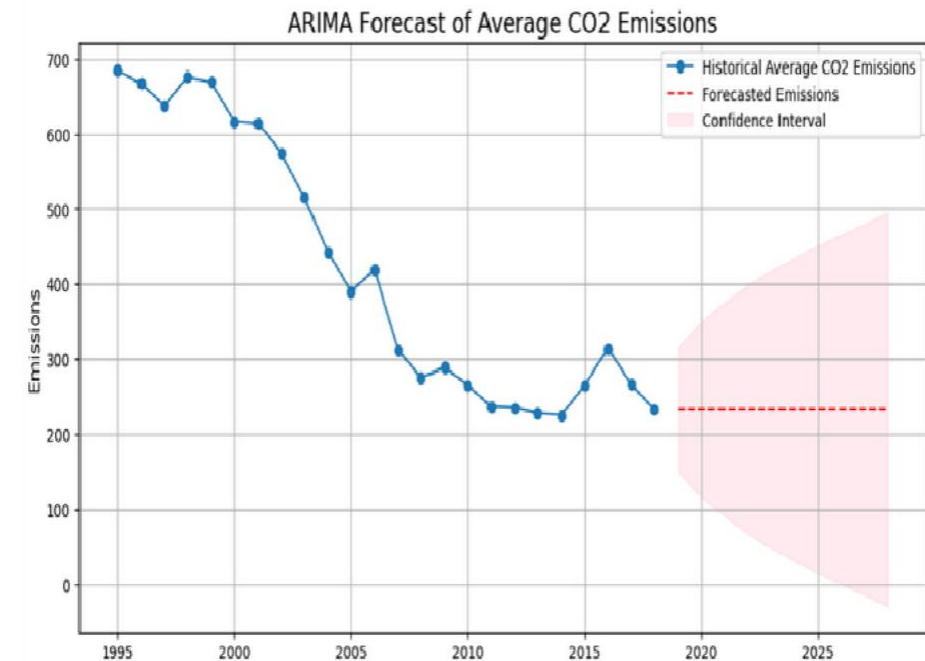
- Aggregated historical CO2 emissions from 1995 to 2018.
- Tested stationarity of the time series using the Augmented Dickey-Fuller (ADF) test.

Forecasting CO2 Emissions Using ARIMA

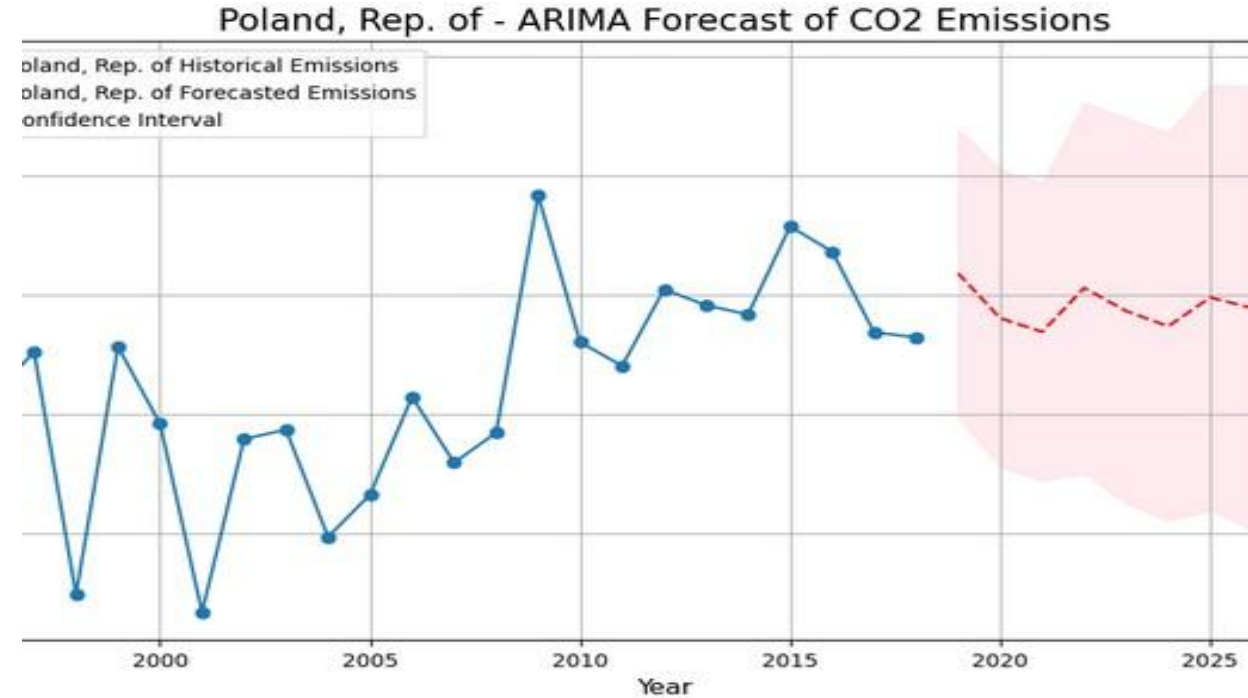
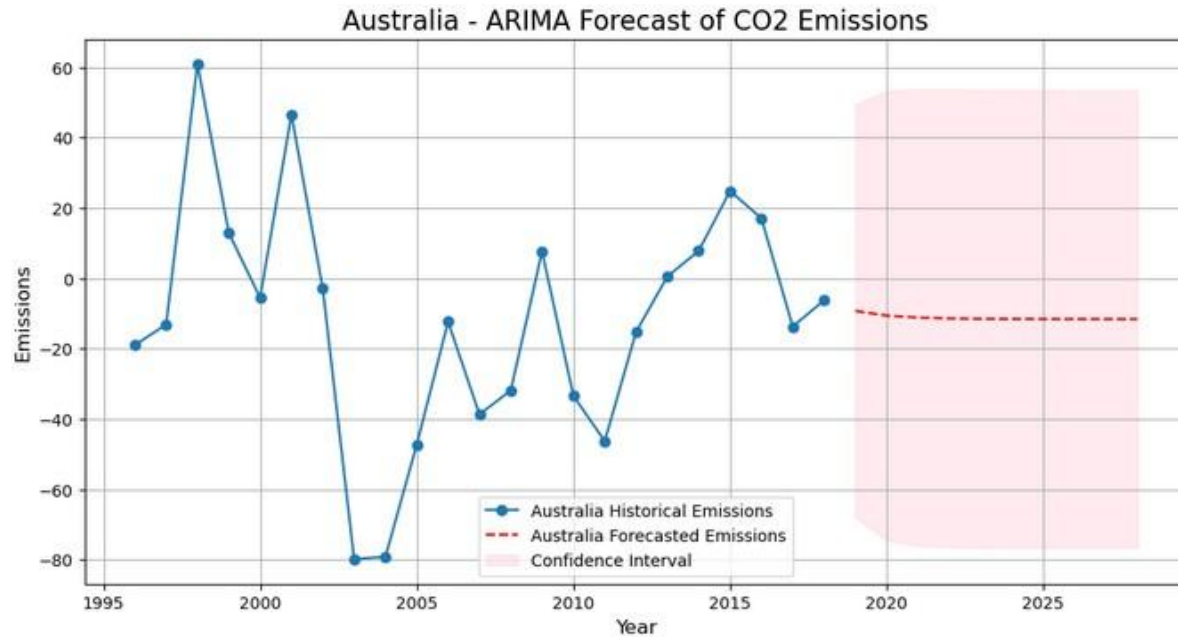
- Applied ARIMA (Auto-Regressive Integrated Moving Average) for forecasting:
 - Auto-ARIMA identified optimal parameters
 - Forecasted for the next 10 years (2019–2028)
- Global average CO2 emissions show a declining trend post-2005
- Forecasted emissions align with recent trends, but uncertainty increases over time (as clearly visible through confidence intervals)
- Visualization includes historical data, forecasted values, and confidence intervals

```
ADF Statistic: -3.2912003419436884
p-value: 0.015276430926989642
The time series is stationary.
Performing stepwise search to minimize aic
ARIMA(2,1,2)(0,0,0)[0] intercept : AIC=240.786, Time=0.10 sec
ARIMA(0,1,0)(0,0,0)[0] intercept : AIC=235.677, Time=0.01 sec
ARIMA(1,1,0)(0,0,0)[0] intercept : AIC=237.561, Time=0.01 sec
ARIMA(0,1,1)(0,0,0)[0] intercept : AIC=237.520, Time=0.02 sec
ARIMA(0,1,0)(0,0,0)[0] intercept : AIC=239.338, Time=0.00 sec
ARIMA(1,1,1)(0,0,0)[0] intercept : AIC=238.513, Time=0.03 sec

Best model: ARIMA(0,1,0)(0,0,0)[0] intercept
Total fit time: 0.185 seconds
Best ARIMA parameters: (0, 1, 0)
```

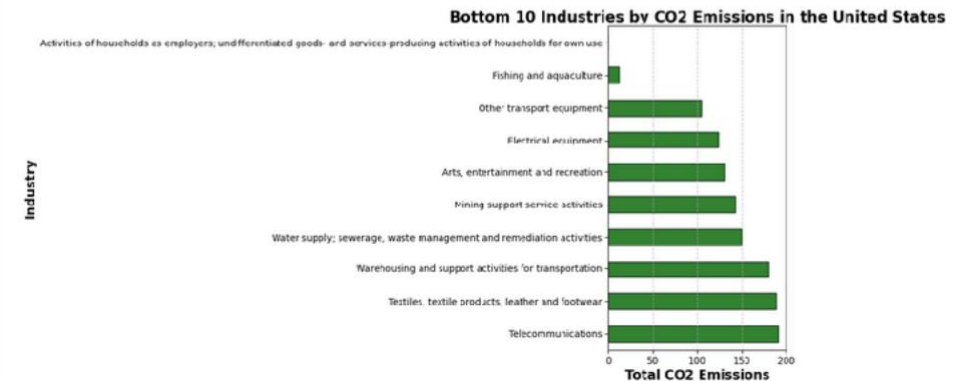
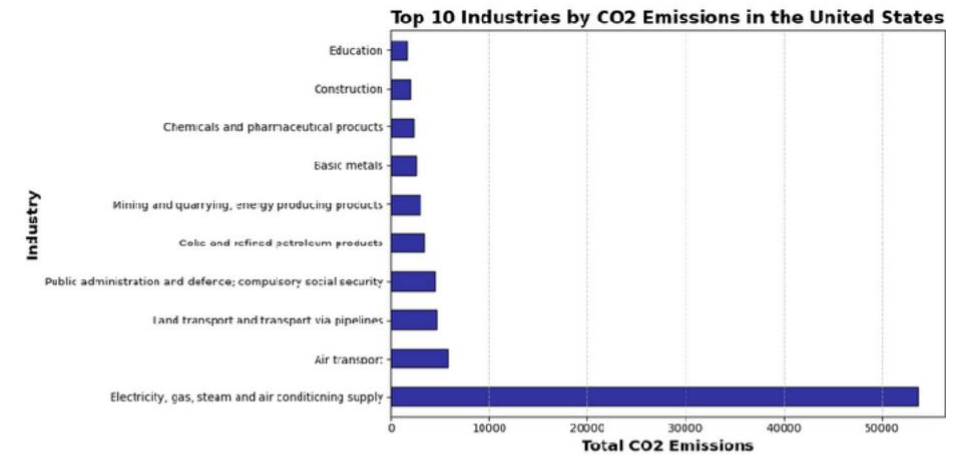
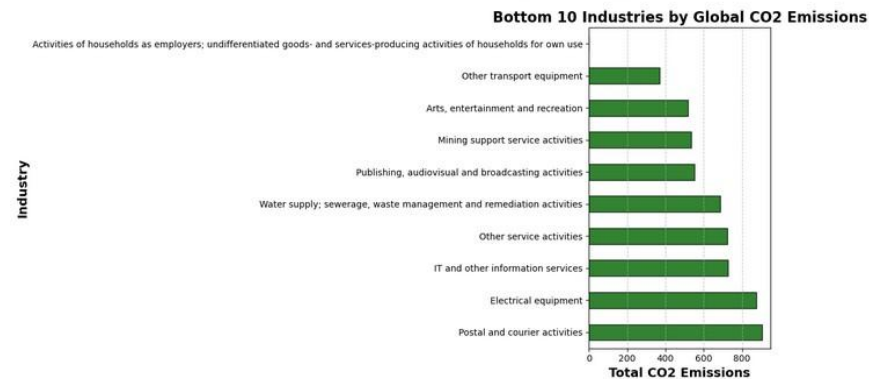
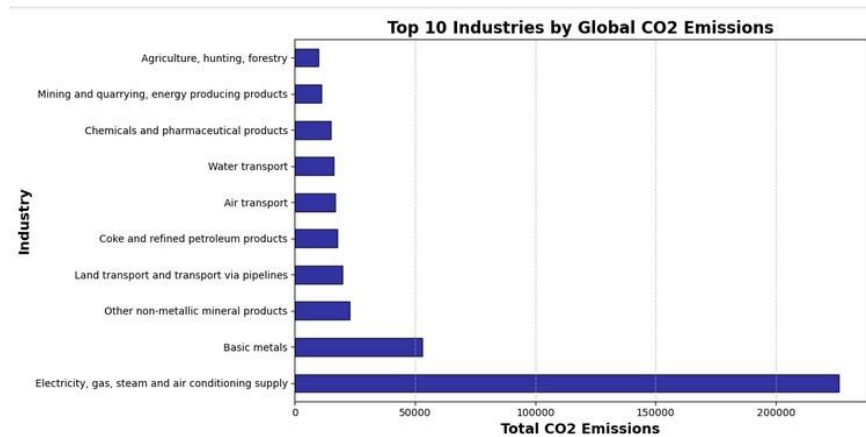


Country-Specific CO2 Emissions Forecast



- Extracted time series data for each country.
- Applied ARIMA modeling to forecast emissions for the next decade.
- **Historical vs. Forecasted Trends:** For example Australia: Moderate and steady patterns with projected flat or declining emissions
- Poland's CO2 emissions exhibit significant **fluctuations**, with periods of sharp increases and decreases
- The forecast indicates a **relatively stable trend** in emissions, with slight oscillations around a central value

Top and Bottom Industries by CO2 Emissions- Globally and In USA



Research Questions

- **What are the historical trends of CO2 emissions by country from 1995 to 2018?**
- **How do CO2 emissions intensities correlate with CO2 emissions across different countries?**
- **Which countries show the most significant potential for reducing CO2 emissions in the future?**
- **How effective are time series forecasting models (e.g., ARIMA) in predicting future CO2 emissions?**
- **What clusters of countries can be identified based on their CO2 emissions trends, and what does this mean for global climate policy?**

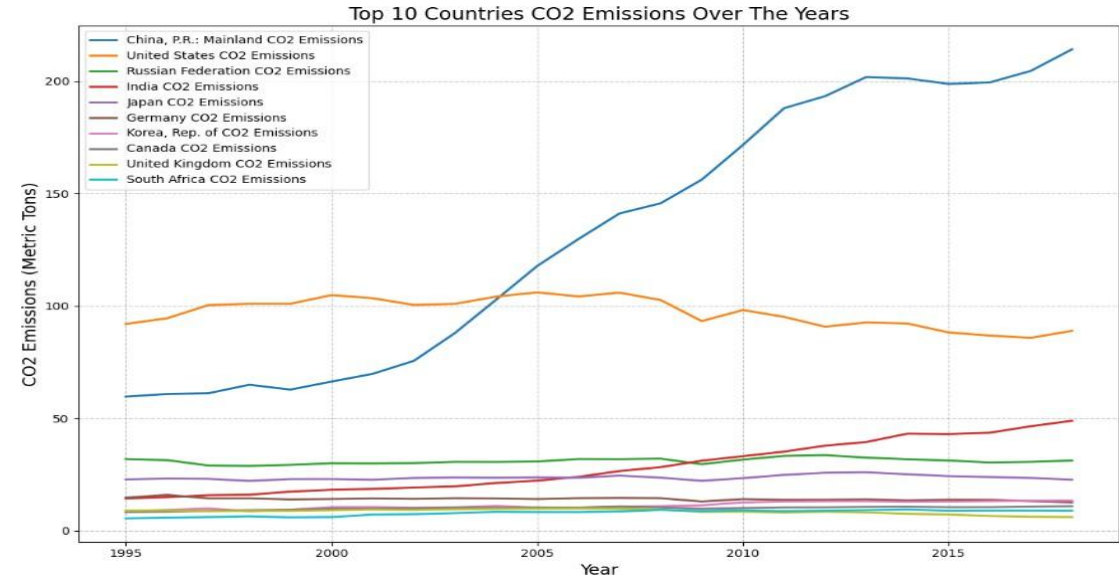
Key Findings

Top emitters: China, U.S., Russian Federation, India.

- Declining trends: UK, Japan.

Forecasts (2019–2028):

- Global emissions: +24% by 2028.
- **Top sectors:** Electricity, gas, steam.



	Year	Forecasted Emissions
0	2019	80735.539885
1	2020	82738.055830
2	2021	84778.345786
3	2022	86857.243478
4	2023	88975.484773
5	2024	91133.814491
6	2025	93332.991329
7	2026	95573.788332
8	2027	97856.993177
9	2028	100183.408446

$$1. 100183.408446 - 80735.539885 = 19447.868561$$

$$2. \frac{19447.868561}{80735.539885} = 0.240892$$

$$3. 0.240892 \times 100 = 24.09\%$$